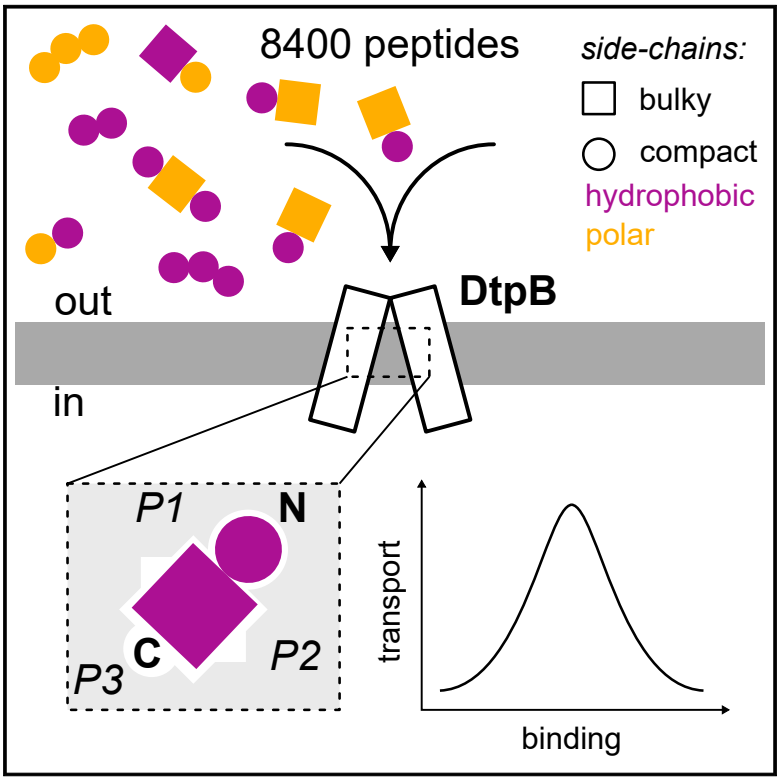
Cell Reports

Article

# Plasticity of the binding pocket in peptide transporters underpins promiscuous substrate recognition

## Graphical abstract




## Highlights

- Structures of peptide transporter DtpB in complex with 14 different di- and tripeptides
- Binding affinities for more than 80 peptides based on thermal shifts
- High-affinity peptides are poorly transported
- Molecular docking of all 8,400 possible di- and tripeptides into DtpB

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## In brief

Kotov et al. determine crystal structures of the peptide transporter DtpB bound to 14 different peptides. This work provides novel insights into the plasticity of the conserved binding pocket. The authors deploy state-of-the-art molecular docking and machine learning to probe whether all 8,400 possible di- and tripeptides bind to DtpB.

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# Plasticity of the binding pocket in peptide transporters underpins promiscuous substrate recognition

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## SUMMARY

Proton-dependent oligopeptide transporters (POTs) are promiscuous transporters of the major facilitator superfamily that constitute the main route of entry for a wide range of dietary peptides and orally administrated peptidomimetic drugs. Given their clinical and pathophysiological relevance, several POT homologs have been studied extensively at the structural and molecular level. However, the molecular basis of recognition and transport of diverse peptide substrates has remained elusive. We present 14 X-ray structures of the bacterial POT DtpB in complex with chemically diverse di- and tripeptides, providing novel insights into the plasticity of the conserved central binding cavity. We analyzed binding affinities for more than 80 peptides and monitored uptake by a fluorescence-based transport assay. To probe whether all 8400 natural di- and tripeptides can bind to DtpB, we employed state-of-the-art molecular docking and machine learning and conclude that peptides with compact hydrophobic residues are the best DtpB binders.

## INTRODUCTION

Living cells must adapt rapidly to environmental changes to maintain nutrient homeostasis, which requires use of different nitrogen-containing nutrients. Hence, cells express a variety of genes to ensure scavenging of alternative nitrogen sources, such as amino acids or peptides.[1] Peptide transporters of the major facilitator superfamily (MFS), the proton-dependent oligopeptide transporter (POT) family, provide the cell with valuable nitrogen and carbon sources by mediating the uptake of di- and tripeptides.[2] POT members are known to have an overall helical arrangement, termed "MFS fold," formed by two helical bundles (N- and C-terminal bundle) that are related by a pseudo-two-fold symmetry[3–8] and function according to the alternate access mechanism.[9–13] Additionally, they carry specific sequence signature motifs important for proton coupling, ligand binding, and transport.[2,3,14] They are known to be highly promiscuous, expected to transport almost all 8,400 di- and tripeptides composed of proteinogenic amino acids.[2]

In *E. coli*, four members of the POT family were identified (YdgR, YhiP, YjdL, and YbgH), which are also known as di- and tripeptide permease (Dtp) A–D, resepectively.[15–17] Experimental structures of DtpA, DtpC, and DtpD have been reported recently.[7,8,18] Transport inhibition experiments have indicated that the binding sites of DtpA and DtpB interact with a large number of substrates and peptidomimetic drugs in a similar fashion as the extensively studied mammalian homolog PepT1.[16,19–24] DtpC and DtpD, however, are classified as atypical POTs because they have been shown previously to favor positively charged peptides as substrates.[18,22,25–30]

In the last decade, over 60 entries of the POT family were deposited in the Protein Data Bank (PDB), representing 11


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bacterial and two mammalian homologs, bound to eight unique natural di- and tripeptides (Ala-Phe, Ala-Glu, Ala-Gln, Ala-Leu, Phe-Ala, Phe-Ala-Gln, Ala-Ala-Ala, and alafosfalin), peptidomimetic drugs (valaciclovir, valganciclovir, and 5-aminolevulinic acid), and a potent transport inhibitor, Lys[Z($NO_2$)]-Val.[3–8,10,18,31–41] These efforts revealed the basic principle underlying peptide binding in POTs, which can be described as an electrostatic clamping mechanism between the invariable part of peptides (N and C termini as well as the peptide backbone) and conserved residues in the transporter (mainly via arginine, lysine, glutamate, asparagine, and tyrosine residues). In addition, the molecular changes during transport have been described recently in human PepT1 and PepT2, providing insights into the dynamics of peptide transporters and the local rearrangements of the binding site throughout the full transport cycle.[31] However, it still remains unclear how POTs can recognize and transport such a vast variety of peptides because of the lack of high-resolution structures of POTs bound to chemically diverse substrates.

Here, we determined crystal structures of DtpB complexed with 14 different di- and tripeptides, providing novel insights into the plasticity of the conserved central binding cavity in response to a wide range of chemically diverse peptides. We thereby also complete the entire family of experimental POT structures from *E. coli* ranging from DtpA to DtpD. Moreover, we measured binding affinities for more than 80 peptides using the thermal shift method and employed a fluorescence-based transport assay to monitor the uptake of peptides into liposomes reconstituted with DtpB. Our analysis indicates that high-affinity peptides are only poorly transported and rather act as inhibitors, while peptides in the medium-affinity range display the highest transport rates. Finally, we employed state-of-the-art molecular docking and machine learning to probe whether all 8,400 di- and tripeptides composed of proteinogenic amino acids can bind to DtpB and conclude that a specific subset of peptides with compact hydrophobic residues are the best DtpB binders.

## RESULTS AND DISCUSSION

### Structures of DtpB-peptide complexes stabilized by nanobody 132

To obtain highly diffracting crystals of DtpB, we selected conformation-specific nanobodies after immunizing a non-inbred llama.[42] Of 31 recombinantly expressed and purified nanobodies, 14 bound DtpB with a $K_D$ of 30 nM or lower, as evident by bilayer interferometry (BLI) measurements (Figures 1 and S1; Tables S1 and S2). They were used as crystallization chaperones in subsequent crystallization trials, and nanobody 132 (Nb132) emerged as the most promising binder for co-crystallization approaches. DtpB-Nb132 was initially incubated with the tripeptide Ala-Leu-Ala (ALA), and well-diffracting crystals grew using the vapor diffusion method. The structure of DtpB was determined using the atomic model of DtpA[7] (PDB: 6GS4) as a search model for molecular replacement. Strong positive peaks in the difference electron density map (i.e., $F_{obs}$-$F_{calc}$) indicated the presence of additional electrons/atoms in the periplasmic region of DtpB and within the central cavity, allowing modeling of Nb132 and the tripeptide.

DtpB crystallized in an inward-facing (IF) open state. In this conformation, the central cavity of the transporter is open to the cytoplasm and closed to the periplasm (Figure 1). DtpB adopts the canonical MFS fold, characterized by two six-transmembrane helix bundles. The N- and C-terminal bundles are linked by the HA-HB helices, as observed in other bacterial POTs.[3–8,10,18,32–35,37,39] The overall structure of DtpB is similar to DtpA, DtpC, and DtpD, with $C_\alpha$ atom RMSD (root-mean-square deviation) values of 0.9, 1.3, and 1.2 Å, respectively. In DtpB, the IF open state is stabilized by electrostatic interactions between the two bundles; i.e., Y31-Y285, G35-Q315, Q49-V440, Y64-Y282, and Y149-E393. Nb132 further stabilizes the closure on the periplasmic side by interacting with the periplasmic surface of the transporter through polar contacts (Figure 1B).

To obtain a broader vision of the plasticity of the binding site in response to peptides possessing various chemical groups, the co-crystallization efforts were continued with an in-house library of 82 peptides. Several batches of the DtpB-Nb132 complex were prepared, incubated with a peptide, and then dispensed robotically in 96-well screens containing three sets of crystallization conditions. When the diffraction resolution of an obtained crystal was better than 4 Å, the chemical screens were further refined around the best conditions. In this campaign, X-ray diffraction data were collected and analyzed for more than 2,000 crystals with the help of automated crystallography pipelines based on the CrystalDirect technology.[43,44] For each dataset with a resolution limit higher than 3.5 Å, the presence or absence of a peptide was initially assessed by using an atomic model of DtpB devoid of any substrate to calculate a difference electron density map (i.e., $F_{obs}$-$F_{calc}$). There, strong positive peaks in the central cavity indicated the presence of a ligand. The co-crystallized peptide was then modeled in the positive peak as a di- or trialanine moiety first and then mutated to its original sequence as the signal improved during the refinement steps. In the final validation, OMIT maps excluding the modeled ligands were calculated for each structure (Figure S2). In summary, we obtained 14 unique peptide bound datasets in a resolution range between 2.0 and 2.8 Å with the following peptides: Ala-Phe (AF), Ala-Ile (AI), Ala-Leu (AL), Ala-Gln (AQ), Ala-Val (AV), Ala-Trp (AW), Lys-Val (KV), Met-Ser (MS), Asn-Val (NV), Ser-Leu (SL), ALA, Ala-Phe-Ala (AFA), Ala-Pro-Phe (APF), and Ala-Trp-Ala (AWA) (Figures 2 and S3; Table 1). This represents a large portfolio of di- and tripeptides for POTs compared with the present literature data. These structures now allowed us to analyze changes and local rearrangements in the binding site of DtpB and shed light on how promiscuity is achieved in this transporter family.

### Plasticity in the peptide binding pocket to accommodate diverse peptides

All DtpB complexes crystallized in the same space group with very similar cell dimensions (Table 1). A superposition of the structures highlights that the N termini of all di- and tripeptides are anchored in a similar manner (Figure 2). The primary amine is steadily hooked by N153, N318, and E393 of DtpB. This triad of residues remains fixed in all structures and is conserved in all prototypical POTs[4] (Figure 2C), with the exception of peptide-histidine transporter 1 (PHT1), where N318 is replaced by an

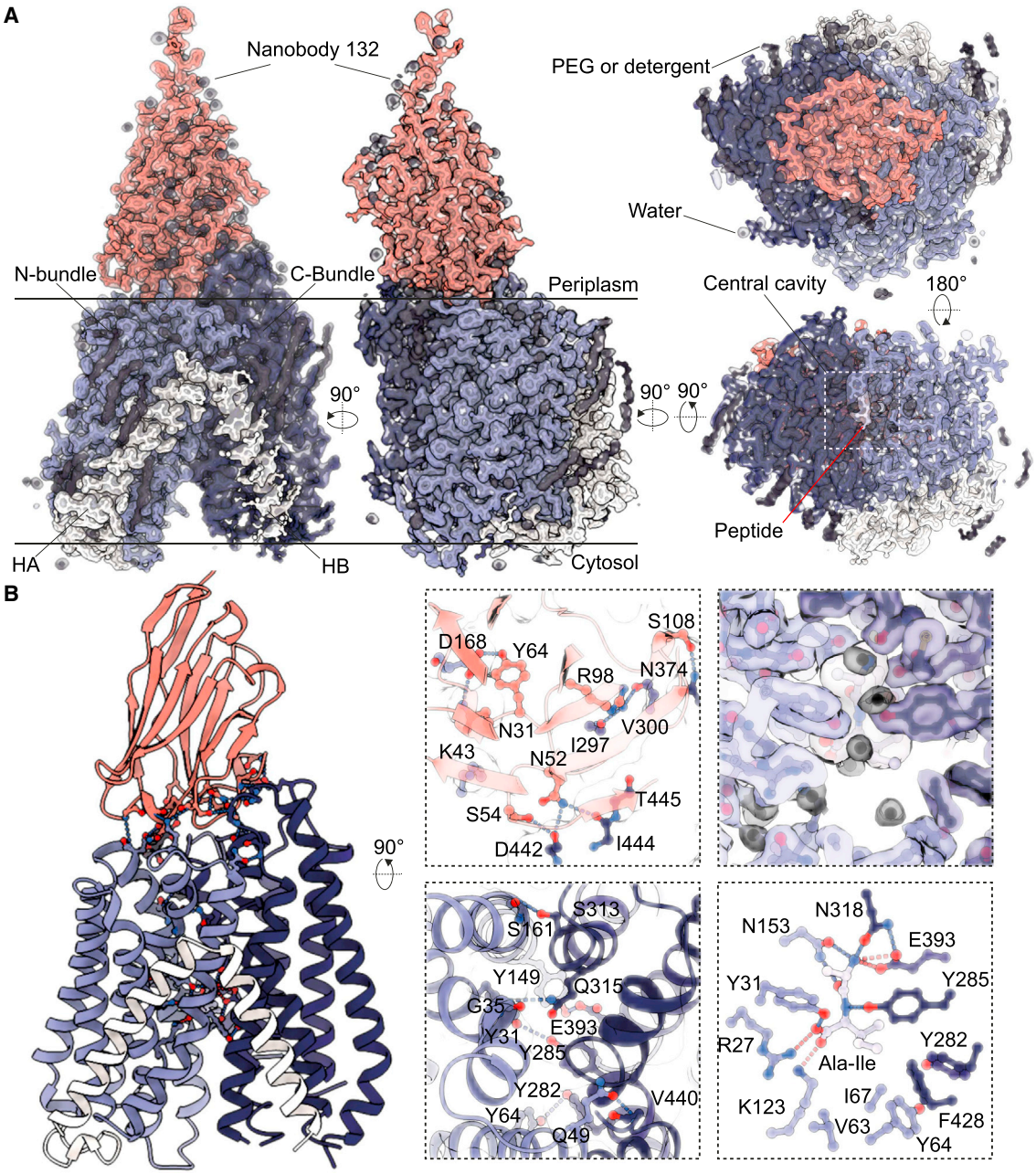



**Figure 1. Structure of DtpB bound to Nb132 and the dipeptide AI**

(A) The atomic model of DtpB-Nb132 derived from the highest resolution dataset with the bound dipeptide AI is presented. The 2Fo-Fc map is shown as a transparent surface (at $\sigma = 1$). The different structural elements are labeled.

(B) Residues stabilizing the observed conformation are displayed as ball and sticks, and the secondary structural elements are shown as ribbons. Interactions between the transporter and the nanobody are shown in the top left close-up view, while interactions stabilizing the IF state between both bundles are highlighted in the bottom panel. The electron density at the binding site and the dipeptide are illustrated in the top right close-up view, and the electrostatic interactions between the peptide and DtpB are shown as dashes in the bottom panel (red dashed lines denote salt bridges; blue dashed lines correspond to polar interactions; waters are shown in black).

aspartate. Additional residues of the binding site form a pocket around the N-terminal residue of the substrate (Figures S3 and S4). These include Y31, Q34, S156, L160, M288, and P319. Together with the conserved N153, N318, E393 triad, they constitute the so-called P1 pocket. The chemical diversity of N-terminal residues in the co-crystallized tripeptide datasets was poor (only alanine residues) but richer for dipeptides (i.e., alanine, lysine, serine, methionine, asparagine). P1 does not

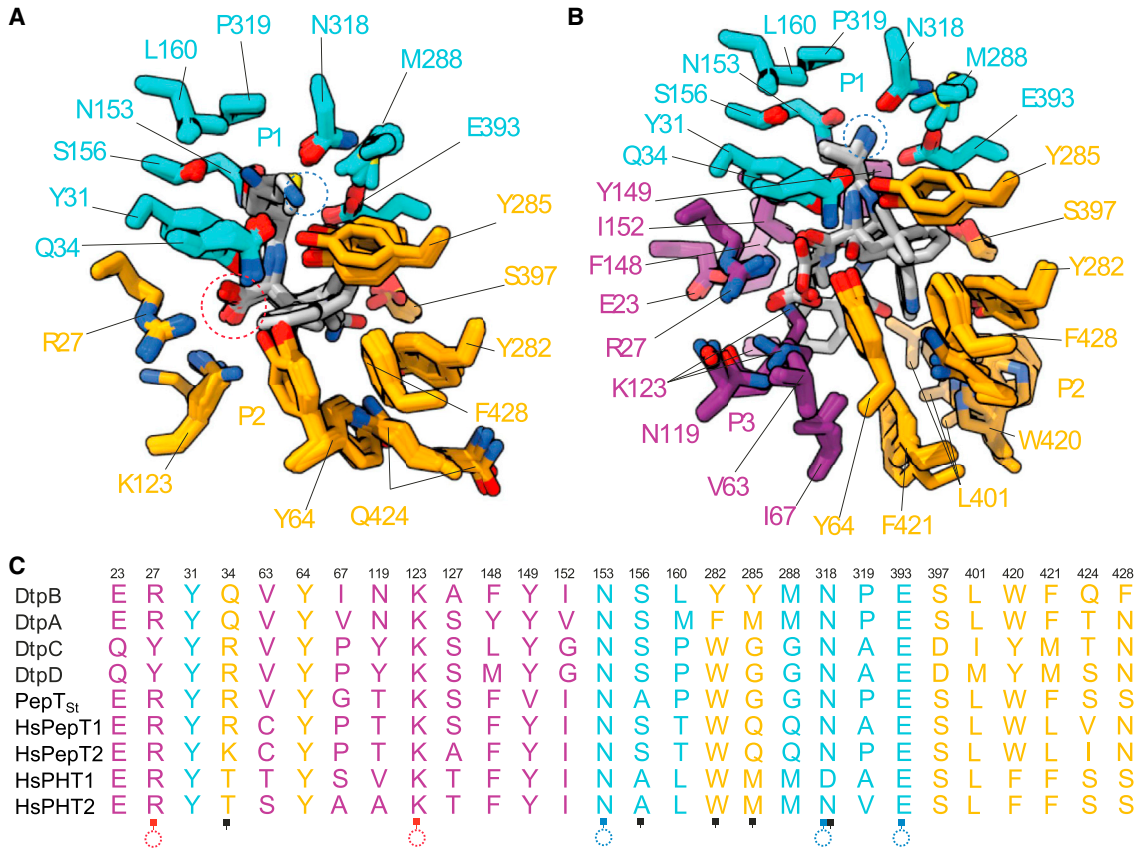



**Figure 2. Definition of the binding pocket of DtpB**

(A) Superimposition of the dipeptide co-crystal structures.

(B) Superimposition of the tripeptide co-crystal structures.

(C) Sequence alignment of the residues constituting the P1, P2, and P3 pockets in POTs. Residues constituting the P1, P2, and P3 pockets are colored in cyan, yellow, and purple, respectively. Blue and red dashed rings circle the N termini and C termini, respectively. Blue and red squares indicate residues mediating electrostatic interactions with the termini of the co-crystallized peptides. Black squares indicate residues mediating polar interactions with the side chains of co-crystallized peptides. The N termini are all coordinated in the same manner, while the C termini adopt different positions in tripeptides.

undergo conformational changes in presence of these various residues; however, polar interactions stabilize certain substrates (Figures 2 and S3; Tables S3 and S4). Notably, Q34 interacts with the ε-amino group of K1* in KV, N318 with the thioester group of M1* in MS, S156 and N318 with the hydroxymethyl group of S1* in SL, and S156 and N318 with the carboxamide group of N1* in NV. The C terminus of dipeptides adopted a constant position, stabilized by R27 and occasionally by K123 as well.

The picture is different for the second residue of the substrate. To adapt the central binding cavity to the different sizes of side chains carried by the C-terminal residues of dipeptides (P2 pocket), two rotamer conformations of Y64 and Q424 are possible (Figures 2A, S3D, and S5). They control the volume of the "upper region" of the P2 pocket. For instance, in the presence of the small dipeptide AV, the upper region of P2 is tightened by the conformation of Y64 and Q424, while it is widened in the presence of the bulky dipeptide AW (Figures 2A and S5; Table S4). Additional residues, such as Y285 and F428, further fine-tune the upper region of P2. Of note, Y285 and Y64 delimit the P2 pocket from the P1 and P3 pockets, respectively (Figure 2). Unlike other side chains fitting in P2, the indole moiety of W2* in the tripeptide AWA extends farther down, toward the cytosolic side of the C bundle (Figure S5). This "lower region" of P2 (i.e., L401, W420, and F421) is more flexible and closes up to stabilize W2*. Polar interactions also occur between P2 and the peptides. Y282 stabilizes the indole ring of W2* in AWA as well as the carboxamide group of Q2* in AQ (Figures S3F and S3J). In addition, the hydrophobic side chains in the second position (i.e., in the peptides AV, AL, AI, AF, AW, ALA, AFA, and AWA) are increasingly stabilized as a function of their size and through contraction of the upper region of P2.

The backbone coordination of tripeptides withstands different torsion angles around amide bonds. Because the primary amine of the N terminus remains hooked in place between N153, N318, and E393, the carboxylic group of the C terminus is subsequently shifted or rotated, resulting in different poses (Figures S3 and S2). For instance, the carboxylic group of AWA coincides with the ones of dipeptides. This co-localization is achieved by a kinked backbone geometry of the tripeptide. AFA and ALA are not kinked but stretched. In comparison, in the dipeptides AL

**Table 1. Data collection and refinement statistics of DtpB-Nb132-peptide datasets**

| Peptide | AF | AFA | AI | AL | ALA | APF | AQ | AV | AW | AWA | KV | MS | NV | SL |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| PDB code | 8B18 | 8B19 | 8B1A | 8B1B | 8B1C | 8B1D | 8B1E | 8B1F | 8B1G | 8B17 | 8B1H | 8B1I | 8B1K | 8B1J |
| Resolution (Å) | 49.90–2.30 (2.38–2.30) | 70.30–2.45 (2.55–2.45) | 51.69–2.15 (2.23–2.15) | 51.95–2.80 (2.90–2.80) | 51.79–2.56 (2.65–2.56) | 62.58–2.30 (2.38–2.30) | 58.74–2.47 (2.56–2.47) | 84.47–2.35 (2.43–2.35) | 59.16–2.50 (2.59–2.50) | 58.76–2.50 (2.59–2.50) | 69.95–2.60 (2.69–2.60) | 50.95–2.55 (2.64–2.55) | 54.14–2.80 (2.90–2.80) | 51.16–2.67 (2.77–2.67) |
| Space group | P 2 21 21 | P 2 21 21 | P 2 21 21 | P 2 21 21 | P 2 21 21 | P 2 21 21 | P 2 21 21 | P 2 21 21 | P 2 21 21 | P 2 21 21 | P 2 21 21 | P 2 21 21 | P 2 21 21 | P 2 21 21 |
| Cell dimensions a, b, c (Å) α, ß, γ (º) | 54.42, 125.07, 168.71 90, 90 90 | 54.70, 125.51, 169.72 90, 90, 90 | 54.32, 125.74, 168.00 90, 90, 90 | 54.59, 125.84, 169.03 90, 90, 90 | 54.40, 123.54, 169.33 90, 90, 90 | 54.76, 125.16, 169.90 90, 90, 90 | 54.47, 125.39, 168.18 90, 90, 90 | 54.65, 126.17, 168.94 90, 90, 90 | 54.61, 126.41, 168.04 90, 90, 90 | 54.60, 125.36, 168.82 90, 90, 90 | 54.34, 125.71, 168.37 90, 90, 90 | 54.18, 124.68, 167.46 90, 90, 90 | 54.14, 124.93, 165.77 90, 90, 90 | 54.49, 125.66, 168.04 90, 90, 90 |
| Total no. of reflections | 667,300 (64,221) | 578,325 (55,600) | 810,303 (71,235) | 378,285 (38,108) | 75,320 (7,396) | 105,697 (10,434) | 83,033 (8,176) | 649,577 (45,053) | 81,675 (7,972) | 80,967 (8,074) | 72,749 (7,106) | 75,797 (7,414) | 57,042 (5,622) | 65,106 (5,161) |
| No. of unique reflections | 52,149 (5,144) | 43,863 (4,296) | 63,584 (6,245) | 29,527 (2,894) | 37,673 (3,699) | 52,854 (5,217) | 42,145 (4,129) | 49,598 (4,863) | 40,878 (4,000) | 40,497 (4,037) | 36,376 (3,553) | 37,899 (3,707) | 28,523 (2,811) | 32,661 (2,628) |
| Completeness (%) | 99.93 (99.94) | 99.90 (99.95) | 99.94 (99.94) | 99.88 (100.00) | 99.89 (99.78) | 99.93 (99.96) | 99.71 (99.61) | 99.88 (99.94) | 99.15 (98.62) | 98.65 (100.00) | 99.92 (100.00) | 99.92 (99.95) | 99.77 (99.93) | 96.74 (77.81) |
| $I/\sigma I$ | 16.79 (1.47) | 16.45 (1.40) | 18.87 (1.48) | 16.30 (1.57) | 16.74 (1.89) | 13.91 (1.24) | 14.56 (1.82) | 15.18 (0.78) | 12.36 (1.05) | 14.29 (1.43) | 13.06 (1.04) | 9.44 (1.30) | 12.01 (0.88) | 12.66 (1.19) |
| Wilson B-factor (Å$^2$) | 53.52 | 71.01 | 50.5 | 80.22 | 66.52 | 57.56 | 61.28 | 59.41 | 69.68 | 69.65 | 71.69 | 54.22 | 77.16 | 80.58 |
| $R_{merge}$ | 0.08998 (1.649) | 0.09384 (2.310) | 0.07757 (1.664) | 0.13730 (2.112) | 0.02248 (0.4068) | 0.02381 (0.5902) | 0.02899 (0.4735) | 0.11130 (3.0180) | 0.01985 (0.6883) | 0.02077 (0.4964) | 0.02634 (0.6764) | 0.04223 (0.5812) | 0.03269 (0.8924) | 0.02147 (0.675) |
| $R_{meas}$ | 0.09399 (1.719) | 0.09789 (2.405) | 0.08102 (1.742) | 0.14350 (2.198) | 0.03179 (0.5752) | 0.03368 (0.8347) | 0.0410 (0.6697) | 0.11610 (3.199) | 0.02807 (0.9734) | 0.02937 (0.702) | 0.03725 (0.9565) | 0.05973 (0.822) | 0.04623 (1.262) | 0.03037 (0.9545) |
| $R_{pim}$ | 0.02673 (0.4834) | 0.02738 (0.6643) | 0.02302 (0.5110) | 0.04087 (0.6025) | 0.02248 (0.4068) | 0.02381 (0.5902) | 0.02899 (0.4735) | 0.03238 (1.0420) | 0.01985 (0.6883) | 0.02077 (0.4964) | 0.02634 (0.6764) | 0.04223 (0.5812) | 0.03269 (0.8924) | 0.02147 (0.6750) |
| CC1/2 | 0.999 (0.648) | 0.999 (0.522) | 0.999 (0.591) | 0.997 (0.531) | 0.999 (0.694) | 1 (0.609) | 0.998 (0.640) | 0.991 (0.285) | 1 (0.432) | 1 (0.643) | 1 (0.488) | 1 (0.545) | 1 (0.367) | 1 (0.350) |
| CC* | 1 (0.887) | 1 (0.828) | 1 (0.862) | 0.999 (0.833) | 1 (0.905) | 1 (0.870) | 1 (0.883) | 0.998 (0.666) | 1 (0.777) | 1 (0.885) | 1 (0.810) | 1 (0.840) | 1 (0.733) | 1 (0.720) |
| Reflections used in refinement | 52,127 (5,142) | 43,856 (4,295) | 63,564 (6,244) | 29,508 (2,894) | 37,667 (3,699) | 52,830 (5,216) | 42,133 (4,122) | 49,579 (4,862) | 40,846 (3,993) | 40,479 (4,037) | 36,360 (3,553) | 37,882 (3,705) | 28,489 (2,809) | 32,590 (2,577) |
| Reflections used for $R_{free}$ | 2,589 (251) | 2,223 (236) | 3,113 (316) | 2,999 (306) | 1,892 (188) | 2,648 (249) | 2,078 (213) | 2,514 (252) | 2,015 (191) | 2,000 (193) | 1,807 (180) | 1,838 (200) | 1,382 (128) | 1,621 (122) |
| $R_{work}$ | 0.2267 (0.3774) | 0.2241 (0.3115) | 0.2154 (0.3490) | 0.2254 (0.3354) | 0.2151 (0.2944) | 0.2299 (0.3515) | 0.2221 (0.3108) | 0.2164 (0.2998) | 0.2245 (0.3442) | 0.2314 (0.3054) | 0.2257 (0.3659) | 0.2241 (0.3319) | 0.2260 (0.3869) | 0.2211 (0.3589) |
| $R_{free}$ | 0.2536 (0.4101) | 0.2673 (0.3354) | 0.2390 (0.3854) | 0.2633 (0.3708) | 0.2410 (0.3293) | 0.2575 (0.3866) | 0.2669 (0.3498) | 0.2363 (0.3244) | 0.2481 (0.3548) | 0.2647 (0.3278) | 0.2602 (0.3766) | 0.2574 (0.3686) | 0.2503 (0.4129) | 0.2636 (0.3964) |

(*Continued on next page*)

**Table 1.** ***Continued***

| Peptide | AF | AFA | AI | AL | ALA | APF | AQ | AV | AW | AWA | KV | MS | NV | SL |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Number of non-hydrogen atoms | 4,718 | 4,793 | 4,787 | 4,669 | 4,781 | 4,792 | 4,786 | 4,785 | 4,786 | 4,791 | 4,787 | 4,784 | 4,789 | 4,787 |
| Protein residues | 580 | 581 | 580 | 580 | 581 | 581 | 580 | 580 | 580 | 581 | 580 | 580 | 580 | 580 |
| RMSD Bond length (Å) | 0.011 | 0.011 | 0.01 | 0.012 | 0.012 | 0.01 | 0.011 | 0.01 | 0.01 | 0.012 | 0.012 | 0.011 | 0.013 | 0.01 |
| RMSD Angle (°) | 1.32 | 1.3 | 1.08 | 1.44 | 1.45 | 1.32 | 1.32 | 1.1 | 1.32 | 1.32 | 1.37 | 1.3 | 1.56 | 1.19 |
| Ramachandran favored (%) | 98.42 | 97.72 | 98.06 | 96.83 | 98.42 | 98.42 | 98.42 | 98.06 | 98.06 | 97.54 | 97.89 | 97.18 | 97.89 | 97.36 |
| Ramachandran allowed (%) | 1.41 | 2.11 | 1.94 | 3.17 | 1.58 | 1.58 | 1.58 | 1.94 | 1.94 | 2.46 | 2.11 | 2.82 | 1.94 | 2.64 |
| Ramachandran outliers (%) | 0.18 | 0.18 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0.18 | 0 |
| Rotamer outliers (%) | 0.42 | 1.06 | 1.06 | 1.91 | 0.64 | 0.64 | 0.64 | 1.06 | 1.7 | 0.85 | 0.64 | 0.85 | 0.64 | 1.06 |
| Clashscore | 10.1 | 10.09 | 8.1 | 10.94 | 10.3 | 8.61 | 7.05 | 7.57 | 9.15 | 9.24 | 9.46 | 9.15 | 11.36 | 10.83 |
| Average B-factor | 67.63 | 80.49 | 60 | 82.94 | 75.22 | 67.08 | 64.85 | 67.4 | 77.25 | 75.84 | 75.99 | 61.75 | 82.31 | 81.74 |

Statistics for the highest-resolution shell are shown in parentheses.

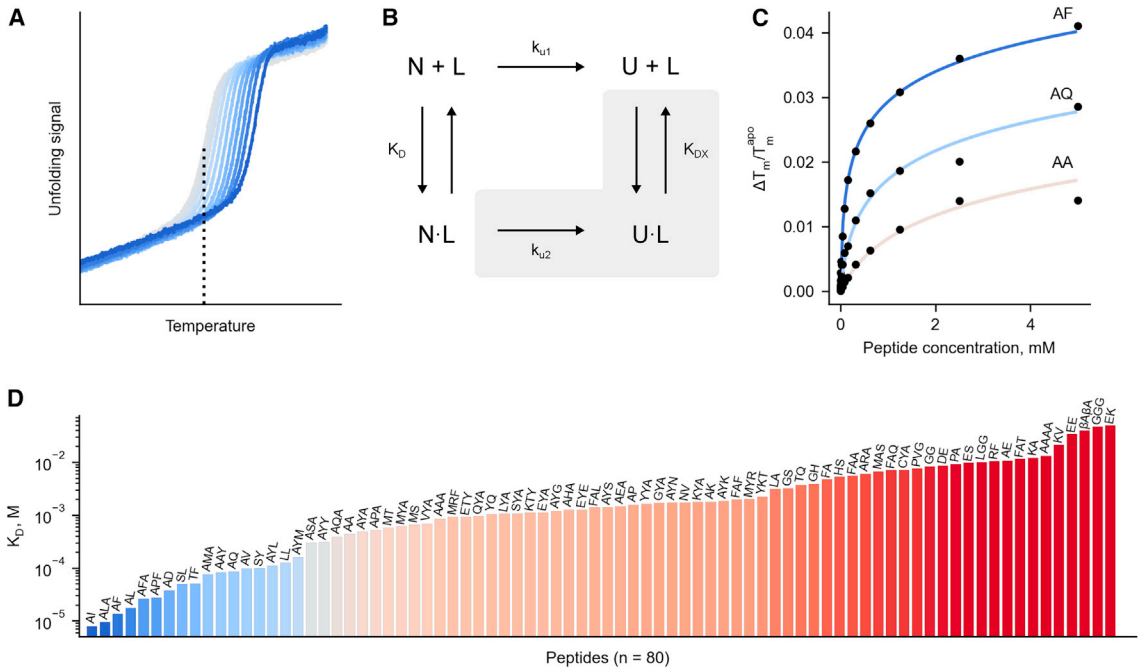



**Figure 3. Large-scale determination of peptide binding to DtpB using a thermal unfolding assay**

(A) Thermal unfolding assays can quantify protein-ligand interactions; ligand concentration increases with blue color intensity. The melting temperature ($T_m$) of the apo state (gray curves) is indicated by a vertical dashed line.

(B) Hall's[55] irreversible unfolding model to describe receptor-ligand interaction in a thermal unfolding assay. N, native state of the receptor; U, unfolded state of the receptor; L, ligand. Elements under the gray area denote the part of the model that was omitted in the current study.

(C) Exemplary saturation curves of diverse peptides. Peptide affinity is color coded, with AF having the highest affinity and AA having the lowest affinity. $T_m^{apo}$ is the $T_m$ of the apo state, and $\Delta T_m$ is the difference in $T_m$ between the apo state in the presence of the respective peptide concentration.

(D) Overview of $K_D$ values obtained in this study. Affinity is color coded, with low-affinity peptides in red and high-affinity peptides in blue.

and AF, a conformational change in K123 creates sufficient space for the carboxylic group of L2* and A3* to fit in a position preserving the stabilizing salt bridges with R27 and K123 (Figures S3 and S6). Finally, in the case of the tripeptide APF, the proline residue restrains the backbone and evicts the C terminus from the positively charged patch formed by R27 and K123, and the bulky phenyl group of F3* extends toward the cytosolic side of the N bundle in P3 (Figures S3 andS6). The overall plasticity of the binding site of DtpB is illustrated in Video S1.

In summary, the N153, N318, E393 triad is a common anchor point of peptide N termini, in agreement with earlier literature.[32,45,46] We find that the C termini of peptides are often stabilized by R27 and K123, but the latter is not mandatory, contrary to the previously suggested model.[14] Recent molecular dynamics (MD) studies using human PepT2 and the tripeptide AAA[36] support this observation and suggest that peptides engage with the binding pocket via A1* first before being tightly locked in place by the triad (N192, N348, E622 in human PepT2). It is likely that the presence of bulky side chains in the N-terminal position would cause local changes in P1, but we did not succeed in determining structures of such complexes. The versatility of P2 has been described previously with rearrangements of Y68 and W427 in a POT from *Streptococcus thermophilus* ($PepT_{St}$),[32,33] which correspond to Y64 and W420 in DtpB. Here, these two residues critically contribute to adapt P2 to the various co-crystallized peptides, but other residues (Y282, Y285, L401, Q424, and F428) are also involved. Except for the different rotamer conformations of K123 to enable various positions of tripeptides, the P3 pocket was rather stable compared with P2. Although these observations indicate that the plasticity of POTs originates mainly in the P2 pocket, many more combinations of peptides are able to bind to and be transported by DtpB and other prototypical POTs. Besides, not all residues involved in ligand promiscuity are conserved in POTs. This could explain the differences of substrate affinities between homologs reported in the literature.

## Determination of peptide affinities using a thermal unfolding assay

To shed further light on the interactions of different di- and tripeptides with DtpB and determine binding affinities for a large set of peptides, we employed a thermal unfolding assay. This approach is commonly used to characterize the stability of proteins and their functional interactions.[47–50] In particular, the stability of proteins (as judged by the melting temperature [$T_m$]) increases in a concentration-dependent manner when a ligand is added (Figure 3A). Various concepts have been proposed to obtain a ligand $K_D$ based on the ligand-induced shifts of $T_m$.[51,52] These approaches typically assume classic thermodynamic behavior of proteins during unfolding; i.e., the equilibrium is quickly reached at all temperatures, and protein unfolding is fully reversible. The latter is rarely observed in practice, so a

kinetic description of the unfolding process should be used instead.[53,54] Recently Hall[55] introduced and validated a kinetic model to determine affinities from thermal shifts (Figure 3B). We modified the Hall[55] model to take advantage of modern non-linear curve fitting methods (STAR Methods) and applied it to DtpB titrated with di- and tripeptides. Titration curves could be fit with high confidence for peptides in a broad affinity range (Figure 3C). The resulting $K_D$ was not strongly affected by assay conditions[56] (Figure S7A). Importantly, our approach allows us to predict the $K_D$ of new peptides obtained from $T_m$ measurements at a single concentration (STAR Methods; Figure S7B). The rank order of the $K_D$ agreed well with an orthogonal technique (microscale thermophoresis [MST]; Figure S7C). With this approach we could efficiently measure $K_D$ values for 82 di- and tripeptides (Figure 3D) to DtpB and cover a broad region of the peptide chemical space. As expected, the determined affinities span several orders of magnitude, with the tightest binders showing affinities in the low micromolar range, while others interacted with DtpB only poorly or not at all.

## Tight peptide binding is not associated with transport

The core biological function of a transporter is its ability to move molecules across the lipid bilayer. To gain more detailed insights into the uptake of various peptides via DtpB, we established a robust transport assay in liposomes, termed "pyranine assay." Because the peptide uptake by POTs is coupled to protons,[57–59] we used the pH-sensitive fluorescent dye pyranine[60] to indirectly monitor peptide transport into DtpB-containing liposomes. Such an approach has been applied recently to characterize the bacterial POT $PepT_{St}$.[61] We utilized similar experimental conditions to follow the transport activity of DtpB reconstituted into liposomes (Figure 4A). To validate the assay, we initially confirmed transport of the known POT substrates Ala-Ala (AA) and Gly-Gly (GG) by DtpB (Figure 4B) and the related transporter $PepT_{St}$.[16,61]

To account for batch-to-batch variations between different liposome reconstitutions, we developed a non-linear curve-fitting procedure to quantify the obtained transport curves in this assay (STAR Methods). In brief, the experimental data were corrected for the empty liposome signal and then fit to a single exponential decay function to obtain the time constant $\tau$ (tau) and the amplitude of the transport curve. The ratio of the amplitude to $\tau$ represents the initial transport rate at time point $t_0$ (slope($t_0$)), corresponding to the time of addition of valinomycin (Figure 4B). Transport measurements of the substrate AA and buffer alone served as a positive and a negative control and were used to normalize slope($t_0$) for all measured peptides to range from 0–1, thus obtaining slope$(t_0)_{rel}$. Quantification of transport rates at varying substrate concentrations allowed us to determine the apparent Michaelis-Menten constants ($K_M$) for the dipeptides AA (0.29 ± 0.04 mM) and GG (2.63 ± 0.65 mM) (Figure S8A). These values are within the expected range for POTs.[10,40] In addition, we used surface electrogenic event reader (SURFE$^2$R)[62] as an orthogonal technique to verify the uptake of GG and AA by DtpB. In this assay an electrochemical gradient is absent, and transport of peptides is driven by an excess of substrate in the external buffer. With this, we could confirm that AA and GG are transported by DtpB; however, the apparent $K_M$ values were 20- to 30-fold higher compared with the electrochemical-gradient-driven conditions as used in the pyranine assay (Figure S8).

Subsequently, we determined slope$(t_0)_{rel}$ for 23 di- and tripeptides using the pyranine assay, covering a broad spectrum of binding affinities as determined by thermal unfolding (Figures 4C and 4D). In the context of available $K_D$ values, slope$(t_0)_{rel}$ forms a bell-shaped distribution (Figure 4D). Peptides that bound poorly to DtpB or not at all in our thermal shift assay exhibited low or no transport. This indicates that peptides with very low binding affinities would not initiate the transport cycle because they do not reside long or well enough bound in the binding site for any conformational changes of the transporter to occur. Alternatively, the correlation of low binding affinities and no apparent transport might also result from clashes or unfavored positioning of the peptide in the binding site. The highest slope$(t_0)_{rel}$ (i.e., peptides with at least 20% of the AA transport rate) could be observed for peptides with $K_D$ values in the range of ~100 μM to ~2.5 mM. To confirm this observation, we measured transport for three more peptides picked from this affinity range: AH ($K_D$ = 0.10 ± 0.04 mM), MT ($K_D$ = 0.57 ± 0.07 mM), and AK ($K_D$ = 1.77 ± 0.09 mM) (Figure 4D, red dots). Of those, only MT was transported, suggesting that a $K_D$ in a specific range is required but not the only determining factor to enable transport by DtpB.

Our findings demonstrate that tightly bound peptides are either transported very slowly or act as inhibitors of transport. Interestingly, previous characterization of $PepT_{St}$ using the pyranine assay[61] demonstrated transport of peptides AF and ALA, which are not transported by DtpB. To confirm the transport-inhibitory effects of tightly binding peptides, we measured uptake of AA in the presence of increasing concentrations of the high-affinity binder AF (Figure 4E). The obtained $IC_{50}$ value for AF is 2.37 ± 0.68 mM, which corresponds to an inhibitor constant ($K_i$) of 0.25 ± 0.08 mM. The $IC_{50}$ value is two orders higher than the previously reported $IC_{50}$ of ~0.027 mM for competition between radiolabeled AA and AF for $PepT_{St}$.[10] This difference can be attributed to the different peptide concentrations used in the assay (2.5 mM of AA used in this work versus radiolabeled AA used at 0.03 mM concentration[10]). We note that, with our assay setup, we did not aim to determine the number of protons being transported along with each peptide, although it has been shown previously for $PepT_{St}$ that this number may vary between di- and tripeptides.[61] Consequently, the amplitude of the signal in the pyranine assay will be higher for peptides that carry more protons, while the time constant $\tau$ is not affected.

## DtpB preferentially binds small hydrophobic peptides

Next, we asked which peptide properties can be predictive of binding to DtpB. No strong correlation (>0.5) could be observed between experimentally determined peptide $K_D$ values (Figure 3D) and basic chemical descriptors, such as molecular weight, polar surface area, or water-octanol partitioning coefficient LogP (Figure S9A). The highest correlation values (0.3–0.4) were observed for descriptors related to the hydrophobicity of the peptide (LogP and hydropathicity on Kyte-Doolittle scale[63]). This property, plotted together with an orthogonal measure such as molecular weight, shows that high-affinity peptides form a potential cluster (Figure S9B). Because our dataset

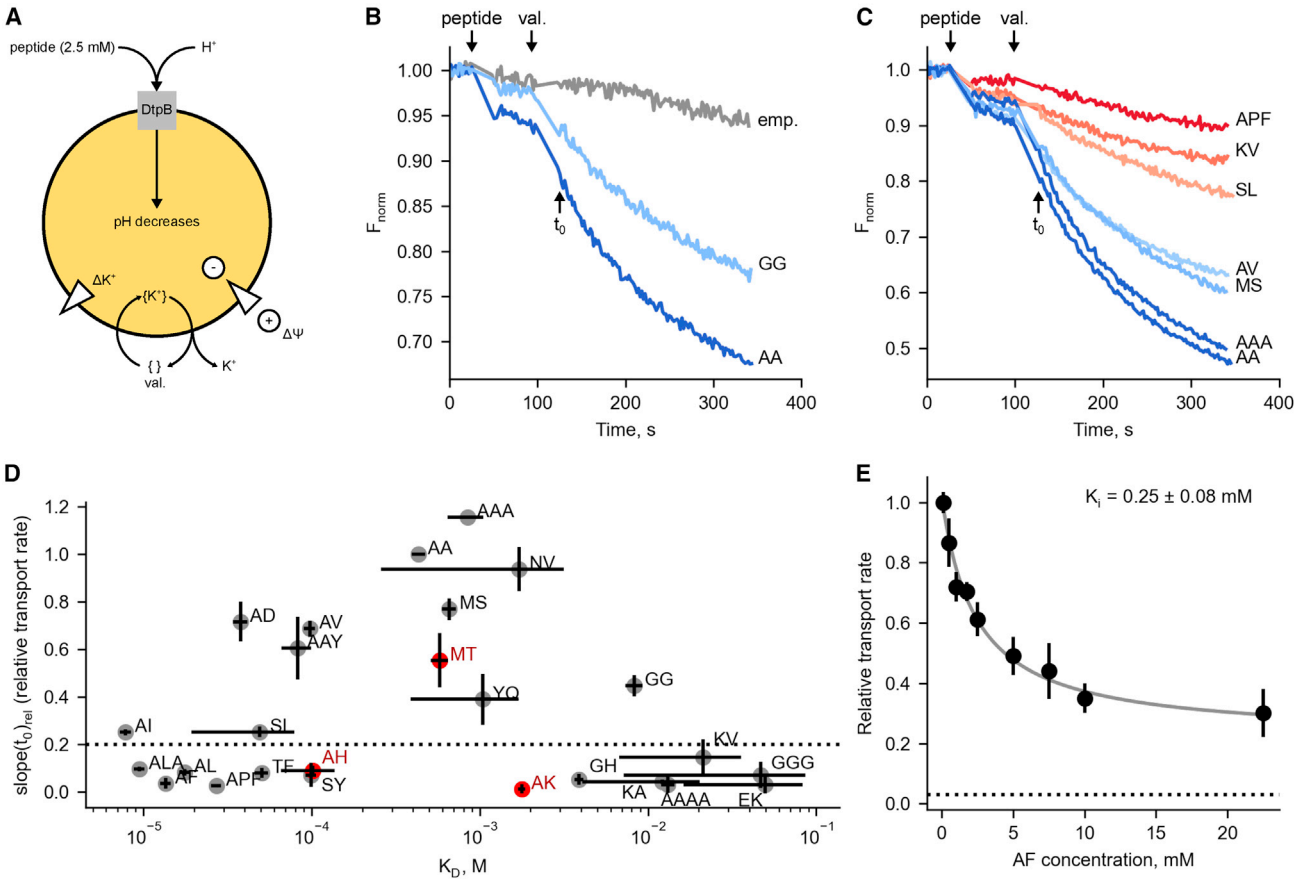



**Figure 4. Measurement of peptide transport by DtpB and its relationship with peptide $K_D$**

(A) The principle of the pyranine assay. DtpB is reconstituted into liposomes, and a concentration gradient of potassium ions ($\Delta K^+$) from the inside to the outside of the liposome is created. Upon addition of valinomycin (val.), the potassium ions are chelated ($\{K^+\}$) and carried across the membrane and establish an electrochemical gradient (membrane potential $\Delta\Psi$). The membrane becomes hyperpolarized, and DtpB can utilize this gradient for proton-coupled peptide transport (the peptide is added at indicated concentration to the outside buffer only). Proton flux into the liposome changes the fluorescence spectrum of the membrane-impermeable dye pyranine (yellow), present in the liposome.

(B) Exemplary transport curves of AA and GG detected with the pyranine assay. The transport signal from liposomes without DtpB (emp.) is shown in gray. The top arrows indicate the approximate time when the peptide or val. was added. Addition of peptide and val. requires the fluorescence reading to be paused. The bottom arrow indicates $t_0$, which corresponds to the time point when the fluorescence readings resumed.

(C) Exemplary transport curves of diverse peptides tested in this work with the pyranine assay. The approximate time when the peptide or val. was added is indicated by arrows. The curves are color coded to show slow or no transport in red and fast transport in blue.

(D) Relationship between binding ($K_D$) and slope$(t_0)_{rel}$ (relative initial transport rate in the pyranine assay). A dashed horizontal line indicates a cutoff of 0.20, which corresponds to 20% of slope$(t_0)$ of AA. Only peptides above the dashed line are considered to be transported. Peptides marked in red were predicted to be transported based on their $K_D$ and subsequently tested in the pyranine assay. Error bars for $K_D$ are standard deviations (n = 2). Error bars for transport rate are the median absolute deviation (n = 2).

(E) AF inhibits transport of AA. Relative transport rates of AA in the presence of variable AF concentrations are shown as black dots, and the fit curve into the Hill equation to determine $IC_{50}$ is shown in gray. Error bars correspond to the standard deviation (n = 3). The relative transport rate of AF alone is shown as a dashed line.

of experimental $K_D$ values accounts for less than 1% of all possible di- and tripeptides made from proteinogenic amino acids (n = 8,400), we asked whether the available crystal structures of DtpB-peptide complexes can be used to formulate more precise peptide recognition rules. For this, we performed flexible docking all of possible di- and tripeptides into DtpB by the Rosetta FlexPepDock protocol[64] and used a modified hit selection procedure (STAR Methods). This allowed us to predict the placement of the peptide inside DtpB for all 8,400 di- and tripeptides (Figures 5A and S10).

The Rosetta energy function is expressed in energy units (kcal/mol) and has been calibrated recently to reliably represent real energies observed in protein molecules.[65] On the other hand, it is impossible to account for all contributing factors that influence ligand binding without a thorough MD simulation. Thus, the scores reported by Rosetta or other docking applications generally show low or no correlation with experimentally observed affinities.[66,67] Indeed, the rank order correlation (Spearman $\rho$) between the Rosetta score for docked peptides and their experimentally measured $K_D$ was only −0.28. Therefore, we performed

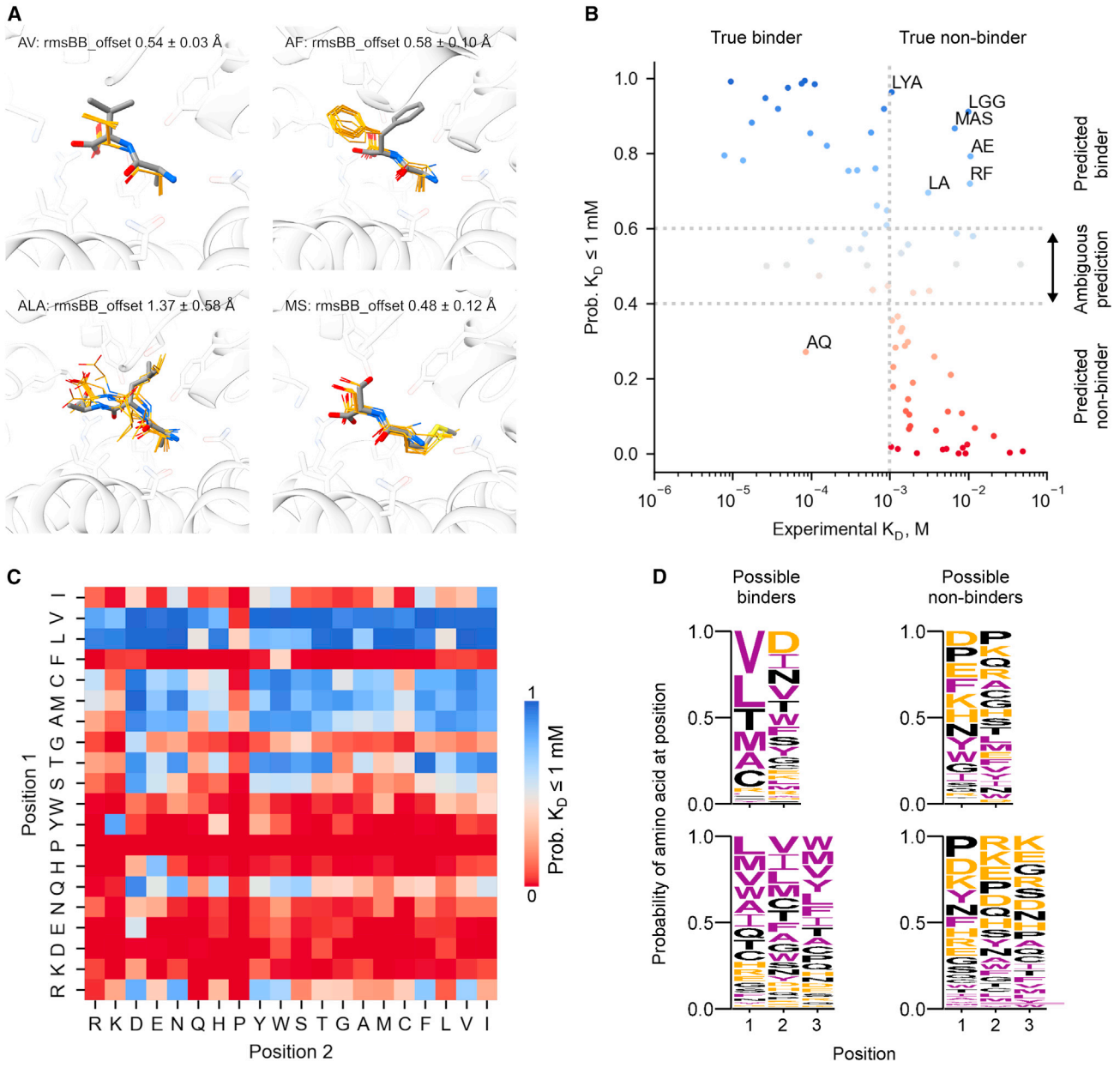



**Figure 5. Flexible docking of di- and tripeptides into DtpB**

(A) Exemplary docking results. The binding pocket is viewed from the cytoplasmic side, and only the native structure of DtpB is shown. The native conformation of the peptide is shown in gray, and the top 10 docked models are shown in orange. For all 14 peptides with known experimental structures, see Figure S10.

(B) Characterization of the logistic regression classifier that predicts binding of a peptide to DtpB. A vertical dashed line shows the value of the experimental $K_D$ that is used to divide peptides into two classes: binders ($K_D \leq 1$ mM) and non-binders ($K_D > 1$ mM). Horizontal dashed lines correspond to the cutoff values for probabilities reported by the logistic regression classifier; if the binder probability is below 0.4, then it is very likely that the peptide will not bind to DtpB. On the other hand, with binder probabilities above 0.6, an interaction with DtpB is to be expected; however, it cannot be excluded that the peptide will still exhibit low affinity. In the probability range between 0.4 and 0.6, it is difficult to assign a peptide to any class ("ambiguous" prediction). Data points are colored by their binder probability (see color map in C). Labeled data points correspond to false positives (top right quadrant) and false negatives (bottom left quadrant).

(C) Influence of amino acid identity on the probability of a dipeptide to be a DtpB binder. Amino acids in columns and rows are ordered by the Kyte-Doolittle hydropathicity scale,[63] with most hydrophobic residues in the top right corner. Note that, in the Kyte-Doolittle scale, W and Y are considered amphiphilic amino acids, so their hydropathicity is close to 0.

(D) Sequence logos (consensus sequence representation) for dipeptides (top row) and tripeptides (bottom row), categorized by their probability to be a binder. Hydrophobic residues are shown in purple, and charged residues are colored orange.

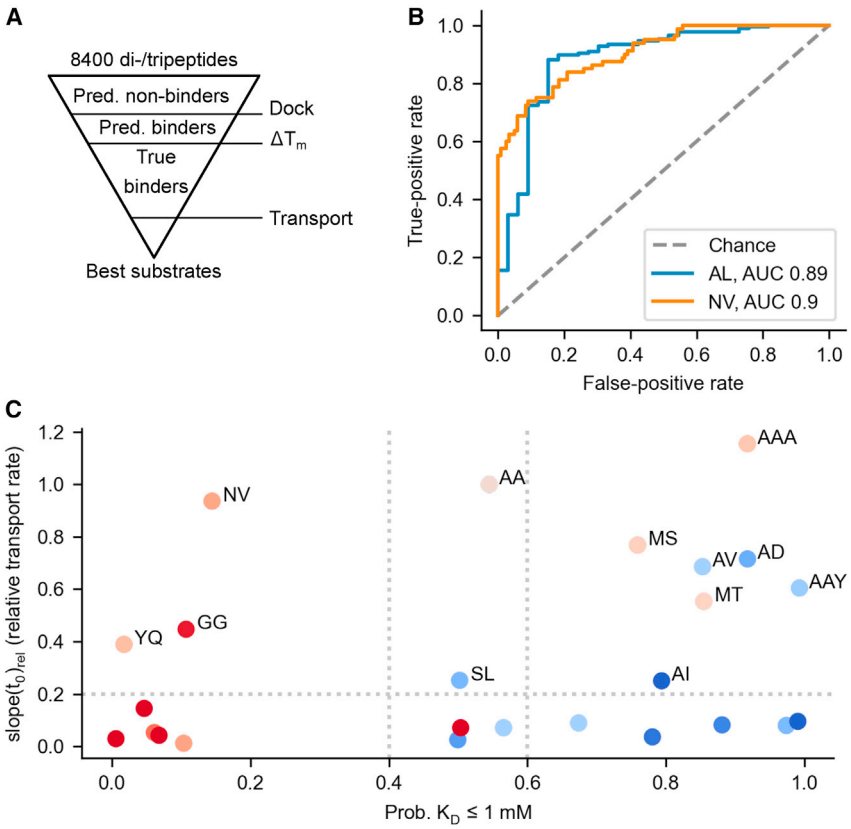



**Figure 6. Characterization of the workflow to discover new peptide substrates for DtpB**

(A) Schematic of the proposed workflow. First, all possible di- and tripeptides (n = 8,400) composed of proteinogenic amino acids are separated into predicted binders and non-binders based on the docking results. This step is virtually instant because the computation has already been performed. Second, true binders are identified using high-throughput $K_D$ estimation based on thermal shifts ($\Delta T_m$). Finally, selected peptides in the optimal $K_D$ range for transport (between ~100 μM and ~2.5 mM) are tested for transport using the low-throughput pyranine assay to identify true substrates.

(B) ROC for the prediction of peptide docking poses for AL and NV.

(C) Peptides transported by DtpB tend to have high binding probability. The data points are color coded according to affinity: high affinity (i.e., low $K_D$) in blue and low affinity (i.e., high $K_D$) in red. Only peptides above the horizontal dashed line are considered transported. Vertical dashed lines indicate the "ambiguous probability" region; i.e., where the binder/non-binder nature of the peptide is predicted with low confidence.

multiple linear regression by using individual Rosetta energy terms (e.g., electrostatic interactions, optimal placement of rotamers[65]) as independent variables (features) and negative decimal logarithm of $K_D$ as the dependent variable. The obtained models performed poorly in cross-validation (CV) tests with a coefficient of determination ($R^2$) score close to 0 or negative. Finally, we separated the di- and tripeptides with experimentally determined $K_D$ into two classes, binders ($K_D \leq 1$ mM) and non-binders ($K_D > 1$ mM), and trained a logistic regression classifier to distinguish these classes based on Rosetta energy terms (Figure 5B). The average area under the curve (AUC) of the receiver-operator characteristic (ROC) during CV of this classifier was 0.67 ± 0.04 (STAR Methods), suggesting that it outperforms random classification (AUC 0.5). The AUC of the classifier trained with full data was 0.88. The particular value of logistic regression is that it provides a probability estimate[68] for each sample to belong to each class (termed "binder probability" in this work). If the peptide's predicted binder probability is less than 0.4, then it is very likely to have an experimental $K_D$ value above 1 mM (Figure 5B). On the other hand, several peptides with a binder probability above 0.6 (AE, LA, LGG, LYA, MAS, and RF) may still be poor binders when tested experimentally. This can be attributed to the fact that the solvent interaction is not modeled, which, for instance, plays an important role in peptide recognition by the periplasmic binding proteins OppA[69] and DppA.[70] Furthermore, "structural snapshots" (as opposed to an MD simulation) are fundamentally limited in capturing the entropic component of binding.[62] Still, the proposed classification approach allows us to exclude obvious non-binders (binder probability below 0.4), which constitute 4,192 of 8,400 peptides (Figure S9B). Only 723 peptides have a binder probability over 0.99; i.e., in the search of new peptide binders to DtpB, it is sufficient to experimentally check ~9% of all possible di- and tripeptides.

Next, we used the binder probability as a proxy to explore the peptide recognition landscape of DtpB (Figures 5C, 5D, and S11). We observe a critical importance of the first position in binding for di- and tripeptides (Figures 5C and S11). In particular polar amino acids, but also bulky hydrophobic amino acids, tend to decrease the binder probability (Figures 5C and S11). Our docking results also support the observation that DtpB preferentially binds hydrophobic peptides (Figure 5D). For instance, a hydrophobic amino acid with a non-bulky side chain (e.g., A, V, L, and M) is found in 67% of predicted binder dipeptides and in 37% of predicted binder tripeptides. Furthermore, the median hydropathicity (Kyte-Doolittle scale) of peptides with a binder probability above 0.8 is 2.2 vs. −1.7 median hydropathicity of all 8,400 possible di- and tripeptides. This agrees with previous data and an earlier study of the yeast homolog Ptr2p[71] and MD-based predictions for $PepT_{St}$.[72] Interestingly, our analysis also reveals that tripeptide binders predominantly contain hydrophobic residues in position 2, whereas dipeptide binders may also contain hydrophilic residues in position 2 (Figure 5D).

## Prediction and validation of peptide pose and binding for DtpB

To make use of the assays and tools described above, we hypothesize that the combination of computational docking prediction tools and experiments can be integrated in a workflow to accelerate identification of DtpB substrates (Figure 6A). First, the

computed docking poses can be used to estimate the binder probability for all possible di- and tripeptides. Second, top hits are validated with thermal unfolding assays to determine the binding affinity, which can be performed experimentally on a medium to high throughput level. Last, the peptides that exhibit an intermediate binding affinity (see the bell-shaped dependence of transport rate vs. $K_D$ in Figure 4D) are analyzed for uptake by a low-throughput transport assay, such as liposome-based or cell-based uptake assays.

The obtained docking model (see DtpB preferentially binds small hydrophobic peptides) was developed with the input of 12 experimental structures of DtpB bound to various peptides determined in this study. The datasets of DtpB bound to dipeptides AL and NV were not part of the training set and therefore can be used to validate the predicted binding poses. The RMSD values of the peptide backbone between the experimental structures and the top 10 docking poses (rmsBB_offset; STAR Methods) were 0.61 ± 0.16 Å for AL and 0.58 ± 0.15 Å for NV (Figures S10 and S12; Table S5). The AUC of the modified hit selection procedure (when applied to all 200 generated dock poses) was 0.89 for AL and 0.9 for AV (Figure 6B), highlighting that the peptide binding pose in the DtpB binding pocket can be predicted with high confidence. Furthermore, the RMSD for the peptide side chains after superposition (rmsSC; STAR Methods) was 1.45 ± 0.07 Å for AL and 1.25 ± 0.24 Å for NV, suggesting that the conformation of the peptide can be predicted with good precision.

To test whether the classifier between binder and non-binder peptides (see DtpB preferentially binds small hydrophobic peptides) can correctly predict binders, we experimentally measured the binding affinity of 12 commercially available peptides that were not used to train the classifier (Table S6; Figure S12). Of those, five peptides (AH, AY, GD, LLA, and RGD) had binder probabilities above 0.5 (predicted binders), and seven peptides (EA, GPE, HH, PY, SH, YA, and YYR) had binder probabilities in the range of 0–0.1 (predicted non-binders). Five of seven non-binders were true negatives (experimental $K_D$ over 1 mM), and two predicted non-binders (PY and SH) turned out to be false negatives with $K_D$ values of 206 ± 126 and 352 ± 38 μM. In case of predicted binders, three of five (AH, AY, and LLA) were true positives (experimental $K_D$ below 1 mM). Two of the false positives among predicted binders (RGD and GD) were in the "ambiguous" probability region (Figure 5B), with binder probabilities of only 0.52 and 0.6 (indicating low confidence of the prediction). The ROC AUC of the experimental validation of the classifier is 0.74; i.e., the discrimination capability of our classification approach is acceptable, but there is room for improvement. If the experimental $K_D$ values from the 12 peptides are added to the training set of the classifier, its final AUC slightly decreases to 0.86. We attribute this result to a smoother (and thus more precise) ROC because of the increased number of samples (n = 81 vs. n = 93). This also indicates that there are some properties of the falsely detected peptides that are underrepresented in the training set and cannot be easily accounted for. Using a larger panel of experimental $K_D$ values (e.g., n = 150 or more) may further improve the performance of the classifier.

Finally, we asked whether selection of peptides using binder probability could exclude the peptides that are known to be transported (see Tight peptide binding is not associated with transport). As demonstrated in Figure 6C, only 3 of 12 peptides transported by DtpB have low binder probability (predicted non-binders). Two peptides, including the reference substrate AA, are in the "ambiguous" probability region, so they are unlikely to be discovered with the proposed workflow; however, the remaining seven known substrates can be potentially identified starting from the docking analysis.

## Conclusions

In this work we present one of the most comprehensive structural and functional characterizations of a POT to date. We established a high-throughput crystallization pipeline and determined the structures of 14 complexes of DtpB with 10 different dipeptides and four tripeptides. Flexible residues within the binding site and multiple stabilizing polar interactions with peptides carrying various chemical groups were identified. Combined with a comprehensive biochemical characterization, these insights allowed us to quantify the binding probability of the whole di- and tripeptide space of proteinogenic amino acids and pinpoint the key properties of a strong binder: high hydrophobicity and moderate size of the side chain in the first position of the peptide.

A common assumption in biochemical studies of transporter proteins is that strong binders are also well transported.[73] For instance, it has been demonstrated for three transporters from the MFS, amino acid-polyamine-organocation (APC) and mitochondrial carrier superfamily (MCS), that stabilizing compounds identified by thermal unfolding assays are also well transported in follow-up transport assays using radioisotope-labeled substrates.[73] In the case of DtpB, however, the trend is different, and only mid-affinity peptides ($K_D$ between 100 μM and 2.5 mM) are well transported in a reconstituted system. We speculate that this could be a general feature of promiscuous transporters as opposed to highly specific transporters. We also note that transport assays are technically more demanding, so development of high-throughput and robust approaches would help advance our understanding of the mechanisms of transport and their relationship with binding. On the other hand, a combination of *in silico* predictions followed by selection of potential interactors using a high-throughput assay can effectively narrow down the number of candidates to be tested in a low-throughput transport assay. In the case of DtpB, applying such a funnel reduces the list of all possible di- and tripeptides (n = 8,400) to a few hundred candidates that can be tested for binding by thermal unfolding assays or other techniques. Next, selected peptides in the medium-affinity range can be tested for transport using liposome-based assays (pyranine assay or radioactive uptake measurements), ultimately identifying new peptide substrates for DtpB and potentially other promiscuous peptide transporters, including the clinically relevant human PepT1, PepT2, PHT1, and PHT2 transporters. A critical step in this application would be to explore the moiety of the putative ligand that can mimic the N terminus of a di- or tripeptide and be used for initial placement of the ligand in the binding pocket. In this work we used a peptide-specific docking protocol; however, a scoring function for small molecules is available in Rosetta,[74] significantly extending the scope of molecules that can

be characterized *in silico*. Given the available structural knowledge of POTs and, more generally, MFS transporters, we envision that large docking campaigns can be set up to discover novel substrates. Furthermore, by leveraging the conformation prediction capabilities of AlphaFold,[75,76] such characterization can be extended to transporters without an experimental structure.

Our findings also indicate that high-affinity peptides believed to be taken up by PepT1 in the human gut could, in fact, act as inhibitors, justifying their use as attractive therapies in inflammatory bowel disease (IBD) and colon cancer.[77–82] Overall, this work provides a solid molecular and biochemical basis for understanding how the structural plasticity of POT's binding site allows uptake of a large diversity of ligands. This constitutes a major step forward toward actual structure-based drug design approaches aiming to inhibit these transport shuttle systems or to hijack them to increase drug absorption.

### Limitations of the study

We would like to point out a number of limitations of this work. First, the spectroscopic method to monitor unfolding of DtpB (thermal unfolding assay) cannot capture the full complexity of protein unfolding. In this work we use a two-state unfolding model, which allows us to obtain high-confidence parameter estimates, but this model significantly simplifies the actual unfolding process. Second, one of the unresolved challenges in this work is estimation of the number of protons that are co-transported with each peptide in the pyranine assay. This limits the possibilities of quantitative analysis of the transport curves (see Tight peptide binding is not associated with transport), so we can mainly perform a binary categorization of peptides (detectable/non-detectable transport). Last, the fundamental limitations of the *in silico* approaches to dock peptides into DtpB and predict their binding properties are the inability to estimate the entropic component of the peptide binding to DtpB and the use of implicit solvent to characterize the energy of the peptide/DtpB interface (for a detailed discussion, see DtpB preferentially binds small hydrophobic peptides). Furthermore, the amount of the experimental data—14 X-ray structures and ~90 binding measurements—is borderline to create a robust logistic regression model. Arguably, an even larger dataset would be necessary to train next-generation predictive models.[83] To simplify reuse of our data to build and test new predictive models, we make our pipeline and the supporting code publicly available under a non-restrictive license.

## STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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  - Data and code availability
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  - Characterization of nanobody binding to DtpB
  - Structure determination of DtpB
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  - Determination of $K_D$ from thermal unfolding curves
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## SUPPLEMENTAL INFORMATION

Supplemental information can be found online at https://doi.org/10.1016/j.celrep.2023.112831.

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## AUTHOR CONTRIBUTIONS

Conceptualization, C.L., M.K., K.E.J.J., V.K., and J.L.; investigation, all authors; software, V.K.; writing – original draft, V.K., C.L., M.K., and K.E.J.J.; writing – review & editing, all authors; visualization, V.K., M.K., and K.E.J.J.; supervision, C.L., M.K., K.E.J.J., J.A.M., and J. Steyaert; funding acquisition, C.L., J.A.M., and J. Steyaert.

## DECLARATION OF INTERESTS

The authors declare no competing interests.

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## STAR★METHODS

### KEY RESOURCES TABLE

| REAGENT or RESOURCE | SOURCE | IDENTIFIER |
|---|---|---|
| Antibodies | | |
| Nanobody 119 | This study | Nb119 |
| Nanobody 120 | This study | Nb120 |
| Nanobody 121 | This study | Nb121 |
| Nanobody 122 | This study | Nb122 |
| Nanobody 123 | This study | Nb123 |
| Nanobody 124 | This study | Nb124 |
| Nanobody 125 | This study | Nb125 |
| Nanobody 126 | This study | Nb126 |
| Nanobody 127 | This study | Nb127 |
| Nanobody 128 | This study | Nb128 |
| Nanobody 129 | This study | Nb129 |
| Nanobody 130 | This study | Nb130 |
| Nanobody 131 | This study | Nb131 |
| Nanobody 132 | This study | Nb132 |
| Nanobody 133 | This study | Nb133 |
| Nanobody 134 | This study | Nb134 |
| Nanobody 135 | This study | Nb135 |
| Nanobody 136 | This study | Nb136 |
| Nanobody 137 | This study | Nb137 |
| Nanobody 138 | This study | Nb138 |
| Nanobody 139 | This study | Nb139 |
| Nanobody 140 | This study | Nb140 |
| Nanobody 141 | This study | Nb141 |
| Nanobody 142 | This study | Nb142 |
| Nanobody 143 | This study | Nb143 |
| Nanobody 144 | This study | Nb144 |
| Nanobody 145 | This study | Nb145 |
| Nanobody 146 | This study | Nb146 |
| Nanobody 147 | This study | Nb147 |
| Nanobody 148 | This study | Nb148 |
| Nanobody 149 | This study | Nb149 |
| Nanobody 150 | This study | Nb150 |
| Nanobody 151 | This study | Nb151 |
| Nanobody 152 | This study | Nb152 |
| Nanobody 153 | This study | Nb153 |
| Nanobody 154 | This study | Nb154 |
| Nanobody 155 | This study | Nb155 |
| Nanobody 156 | This study | Nb156 |
| Nanobody 157 | This study | Nb157 |
| Nanobody 158 | This study | Nb158 |
| Nanobody 159 | This study | Nb159 |
| Bacterial and virus strains | | |
| *E. coli* C41(DE3) | Sigma | CMC0017 |
| *E. coli* WK6 | Steyaert lab | N/A |

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*Continued*

| REAGENT or RESOURCE | SOURCE | IDENTIFIER |
|---|---|---|
| Biological samples | | |
| N/A | N/A | N/A |
| Chemicals, peptides, and recombinant proteins | | |
| n-Dodecyl-β-Maltoside (DDM) | Anatrace | D310S |
| n-Decyl-β-D-Maltoside (DM) | Anatrace | D322LA |
| MemGold2 crystallization screen | Molecular Dimensions | MD1-64 |
| Ala-Ala (AA) | Sigma | A502 |
| Ala-Ala-Ala (AAA) | Bachem | H-1445 |
| Ala-Ala-Ala-Ala (AAAA) | Bachem | H1260 |
| Ala-Ala-Tyr (AAY) | Bachem | G-1445 |
| Ala-Asp (AD) | Bachem | G1195 |
| Ala-Glu (AE) | Bachem | 4002318 |
| Ala-Glu-Ala (AEA) | GL Biochem | Custom synthesis |
| Ala-Phe (AF) | Bachem | A3128 |
| Ala-Phe-Ala (AFA) | Bachem | H-5420 |
| Ala-His-Ala (AHA) | GL Biochem | Custom synthesis |
| Ala-Ile (AI) | Bachem | G1260 |
| Ala-Leu (AL) | Bachem | A1878 |
| Ala-Leu-Ala (ALA) | Bachem | H-5975 |
| Ala-Lys (AK) | Bachem | G-1290 |
| Ala-Met-Ala (AMA) | GL Biochem | Custom synthesis |
| Ala-Pro (AP) | Bachem | G1350 |
| Ala-Pro-Ala (APA) | Bachem | H1595 |
| Ala-Pro-Phe (APF) | Bachem | H1615 |
| Ala-Gln (AQ) | Sigma | G8541 |
| Ala-Gln-Ala (AQA) | GL Biochem | Custom synthesis |
| Ala-Arg-Ala (ARA) | GL Biochem | Custom synthesis |
| Ala-Ser-Ala (ASA) | GL Biochem | Custom synthesis |
| Ala-Val (AV) | Bachem | G1320 |
| Ala-Tyr-Ala (AYA) | GL Biochem | Custom synthesis |
| Ala-Tyr-Gly (AYG) | GL Biochem | Custom synthesis |
| Ala-Tyr-Lys (AYK) | GL Biochem | Custom synthesis |
| Ala-Tyr-Leu (AYL) | GL Biochem | Custom synthesis |
| Ala-Tyr-Met (AYM) | GL Biochem | Custom synthesis |
| Ala-Tyr-Asn (AYN) | GL Biochem | Custom synthesis |
| Ala-Tyr-Ser (AYS) | GL Biochem | Custom synthesis |
| Ala-Tyr-Tyr (AYY) | GL Biochem | Custom synthesis |
| Cys-Tyr-Ala (CYA) | GL Biochem | Custom synthesis |
| Asp-Glu (DE) | Sigma | A1916 |
| Glu-Glu (EE) | Sigma | G3640 |
| Glu-Lys (EK) | Bachem | G1955 |
| Glu-Ser (ES) | Bachem | G1980 |
| Glu-Thr-Tyr (ETY) | Bachem | H3210 |
| Glu-Tyr-Ala (EYA) | GL Biochem | Custom synthesis |
| Glu-Tyr-Glu (EYE) | Bachem | H3225 |
| Phe-Ala (FA) | Bachem | G-2850 |
| Phe-Ala-Ala (FAA) | GL Biochem | Custom synthesis |
| Phe-Ala-Asp (FAD) | GL Biochem | Custom synthesis |
| Phe-Ala-Phe (FAF) | GL Biochem | Custom synthesis |

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| REAGENT or RESOURCE | SOURCE | IDENTIFIER |
|---|---|---|
| Phe-Ala-Leu (FAL) | GL Biochem | Custom synthesis |
| Phe-Ala-Gln (FAQ) | GL Biochem | Custom synthesis |
| Phe-Ala-Thr (FAT) | GL Biochem | Custom synthesis |
| Gly-Gly (GG) | Sigma | 50199 |
| Gly-Gly-Gly (GGG) | Sigma | G1377 |
| Gly-Gly-His (GGH) | Sigma | G4541 |
| Gly-His (GH) | Sigma | G1627 |
| Gly-Ser (GS) | Sigma | G3127 |
| Gly-Tyr-Ala (GYA) | GL Biochem | Custom synthesis |
| His-Ser (HS) | Sigma | H3129 |
| Lys-Ala (KA) | Bachem | G-2630 |
| Lys-Thr-Tyr (KTY) | Bachem | H4245 |
| Lys-Val (KV) | Bachem | G2705 |
| Lys-Tyr-Ala (KYA) | GL Biochem | Custom synthesis |
| Leu-Ala (LA) | Bachem | G2460 |
| Leu-Gly-Gly (LGG) | Sigma | L9750 |
| Leu-Leu (LL) | Bachem | M1535 |
| Leu-Tyr-Ala (LYA) | GL Biochem | Custom synthesis |
| Met-Ala-Ser (MAS) | Sigma | M1004 |
| Met-Arg-Phe (MRF) | Bachem | H2965 |
| Met-Ser (MS) | Sigma | M9380 |
| Met-Thr (MT) | Bachem | G2810 |
| Met-Tyr-Ala (MYA) | GL Biochem | Custom synthesis |
| Met-Tyr-Arg (MYR) | GL Biochem | Custom synthesis |
| Asn-Val (NV) | Bachem | G3720 |
| Pro-Ala (PA) | Bachem | G2985 |
| Pro-His-Tyr (PHY) | Bachem | H4755 |
| Pro-Val-Gly (PVG) | Bachem | H4820 |
| Asn-Tyr-Ala (QYA) | GL Biochem | Custom synthesis |
| Arg-Phe (RF) | Bachem | |
| Ser-Leu (SL) | Bachem | G3200 |
| Ser-Tyr (SY) | Bachem | G3240 |
| Ser-Tyr-Ala (SYA) | GL Biochem | Custom synthesis |
| Thr-Phe (TF) | Bachem | G3295 |
| Thr-Gln (TQ) | Sigma | T3275 |
| Val-Tyr-Ala (VYA) | GL Biochem | Custom synthesis |
| Tyr-Lys-Thr (YKT) | Bachem | H5175 |
| Tyr-Gln (YQ) | Bachem | G400 |
| Tyr-Tyr-Ala (YYA) | GL Biochem | Custom synthesis |
| β-Ala- β-Ala (βAβA) | Bachem | G1150 |
| 16:0-18:1 PE (POPE) | Avanti polar lipids | 850757P |
| 16:0-18:1 PG (POPG) | Avanti polar lipids | 840457P |
| trisodium 8-hydroxypyrene-1,3,6-trisulfonate (pyranine) | Sigma Aldrich | H1529 |
| valinomycin | Sigma Aldrich | V0627 |
| **Deposited data** | | |
| Coordinates and structure factors of DtpB+Nb132+AF | This study | PDB: 8B18 |
| Coordinates and structure factors of DtpB+Nb132+AFA | This study | PDB: 8B19 |
| Coordinates and structure factors of DtpB+Nb132+AI | This study | PDB: 8B1A |
| Coordinates and structure factors of DtpB+Nb132+AL | This study | PDB: 8B1B |

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| REAGENT or RESOURCE | SOURCE | IDENTIFIER |
|---|---|---|
| Coordinates and structure factors of DtpB+Nb132+ALA | This study | PDB: 8B1C |
| Coordinates and structure factors of DtpB+Nb132+APF | This study | PDB: 8B1D |
| Coordinates and structure factors of DtpB+Nb132+AQ | This study | PDB: 8B1E |
| Coordinates and structure factors of DtpB+Nb132+AV | This study | PDB: 8B1F |
| Coordinates and structure factors of DtpB+Nb132+AW | This study | PDB: 8B1G |
| Coordinates and structure factors of DtpB+Nb132+AWA | This study | PDB: 8B17 |
| Coordinates and structure factors of DtpB+Nb132+KV | This study | PDB: 8B1H |
| Coordinates and structure factors of DtpB+Nb132+MS | This study | PDB: 8B1I |
| Coordinates and structure factors of DtpB+Nb132+SL | This study | PDB: 8B1J |
| Coordinates and structure factors of DtpB+Nb132+KV | This study | PDB: 8B1K |
| Oligonucleotides | | |
| N/A | N/A | N/A |
| Recombinant DNA | | |
| pNIC-CTHF | Opher Gileadi, Addgene | RRID:Addgene_26105 |
| pMESy4 | Steyaert lab | N/A |
| Software and algorithms | | |
| PR.ThermControl | Nanotemper | v. 2.1.2 |
| MoltenProt | Kotov et al.[84] | v. 0.1.1 |
| symfit | Roelfs and Kroon[85] | v. 0.5.3 |
| GraphPad Prism | Dotmatics | v. 9.4.1 |
| Rosetta FlexPepDock | Raveh et al.[65] | v. 3.12 |
| Rosetta BuildPeptide | Cock et al.[86] | v. 3.12 |
| Biopython PDB module | Hamelryck and Manderick[87] | v. 1.78 |
| Rosetta relax | Nivon et al.[88] | v. 3.12 |
| GNU parallel | Tange[89] | v. 20180922 |
| gemmi | Wojdyr[90] | v. 0.4.3 |
| scikit-learn | Pedregosa et al.[91] | v. 0.24.2 |
| Python | N/A | version 3.7.10 |
| Pandas | McKinney[92] | v. 1.2.4 |
| numpy | Oliphant[93] | v. 1.20.2 |
| scipy | Jones et al.[94] | v. 1.6.2 |
| Octet Data Analysis HT software | Pall ForteBio | v. 10.0.3.7 |
| XDS | Kabsch[95] | https://xds.mr.mpg.de/phaser |
| Phaser | CCP4i2[96] | https://www.ccp4.ac.uk/ |
| Coot | Emsley et al.[97] | https://www2.mrc-lmb.cam.ac.uk/personal/pemsley/coot/ |
| PHENIX | Adams et al.[98] | https://phenix-online.org/ |
| ISOLDE | Croll[99] | https://isolde.cimr.cam.ac.uk/ |
| ChimeraX | Goddard et al.[100] | https://www.cgl.ucsf.edu/chimerax/ |
| WebLogo 31 | Crooks et al.[101] | v. 3.7.8 |
| MO.Affinity | Nanotemper | v. 2.3 |
| Other | | |
| Ni-NTA Agarose | Invitrogen | R901-15 |
| HiLoad 16/600 Superdex 200 column | GE Healthcare | N/A |
| Spin-X® UF 20 mL 50 kDa cut-off concentrator | Corning | 431490 |
| CaptureSelect beads for EPEA-tag | Thermo Fischer Scientific | 194288010 |
| Anti-penta-His sensors | Satorius | 18–5120 |
| Prometheus NT.48 Standard Capillaries | Nanotemper | PR-C002 |

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*Continued*

| REAGENT or RESOURCE | SOURCE | IDENTIFIER |
|---|---|---|
| Monolith NT.LabelFree Standard Capillaries | Nanotemper | MO-Z022 |
| Microspin G-25 spin column | Cytiva | 27532501 |
| 96-well black chimney microplate | Greiner | 655209 |
| SURFE2R N1 Single Sensor 3 mm | Nanion | 16 1001 |
| Code for shared-parameter fitting of titration | This study | https://dx.doi.org/10.5281/zenodo.7611944 |
| Code for pyranine assay curve fitting, data manipulation and visualization | This study | https://dx.doi.org/10.5281/zenodo.7612027 |
| Code and data for docking-based prediction of binder probability | This study | https://dx.doi.org/10.5281/zenodo.7586704 |

## RESOURCE AVAILABILITY

### Lead contact

Further information and requests for resources and reagent should be directed to and will be fulfilled by the lead contact, Christian Löw (christian.loew@embl-hamburg.de).

### Materials availability

Plasmids generated in this study are available from the lead contact upon request without any restrictions.

### Data and code availability

- The structures of the DtpB-peptide complexes have been deposited to the PDB and are publicly available as of the date of publication. Accession numbers are listed in the key resources table.
- All original code for the pyranine assay, thermal unfolding assay and docking-based prediction of binder probability were deposited to Zenodo. DOIs are listed in the key resources table.
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

## METHOD DETAILS

### Protein expression and purification of DtpB

The full-length *dtpb* (Uniprot: P36837) gene was amplified from the *Escherichia coli* genome and cloned into a pNIC-CTHF vector.[102] The construct has a C-terminal $His_6$-tag with a Tobacco Etch Virus (TEV) cleavage site. DtpB was expressed and purified as previously described.[7,103] Briefly, DtpB was overexpressed in *E. coli* C41(DE3) cells in TB medium. The cell pellet was resuspended in lysis buffer (20 mM NaP, pH 7.5, 300 mM NaCl, 5% glycerol, 15 mM imidazole) with 5 units/ml DNase I, 1 tablet of cOmplete EDTA-free protease inhibitor (Roche)/100 mL buffer, 1 mg/mL lysozyme and 0.5 mM TCEP followed by cell lysis with an emulsifier (EmulsiFlex-C3, Avestin, three passages). The lysate was centrifuged for 10 min at 10,000 g and the supernatant centrifuged for 50 min at 95,000 g (Optima XE-90, Beckman Coulter). The pellet containing the membrane fraction was solubilized in 1% n-Dodecyl-β-Maltoside (DDM; Anatrace, sol-grade) for 1 h. The sample was centrifuged for another 50 min at 70,000 g and the supernatant applied to Ni-NTA Agarose beads (Invitrogen). After 60 min incubation on a rotating wheel, the suspension was transferred to a gravity column. Following two wash steps with lysis buffer supplied with 0.5 mM TCEP, 0.03% DDM and 30 mM imidazole, DtpB was eluted with lysis buffer containing 0.5 mM TCEP, 0.03% DDM and 250 mM imidazole. TEV cleavage was performed overnight during dialysis in gel filtration buffer (20 mM HEPES, 150 mM NaCl, 0.5 mM DTT, 5% glycerol and 0.03% DDM), and the protein was further purified by negative IMAC. The flow through was then concentrated and loaded on a HiLoad 16/600 Superdex 200 column (GE Healthcare). All steps following cell lysis were performed at 4°C. The protein was concentrated to 6–8 mg/mL using a 20 mL 50 kDa cut-off concentrator (Corning Spin-X UF concentrators), flash-frozen and stored at −80°C until further use.

### Selection and purification of nanobodies

41 nanobodies representing 28 different nanobody families were generated against purified DtpB following established protocols.[42] Nanobody constructs contain a C-terminal $His_6$-and EPEA-tag for affinity purification (pMESy4 vector). In total, 30 nanobodies representing the different nanobody families were expressed in *E. coli* WK6 cells and purified following standard procedures.[42,104] In brief, the cell pellet was resuspended in TES buffer (0.2 M TRIS, pH 8, 0.5 mM EDTA, 0.5 M sucrose) supplemented with one protease inhibitor tablet and 5 units/ml DNase. An osmotic shock was performed by the addition of diluted TES buffer to release the periplasmic proteins. The solution was first centrifuged for 20 min at 10,000 g and additionally for 30 min at 40,000 g. The supernatant was

applied to CaptureSelect beads (Thermo Fisher Scientific), which were equilibrated with wash buffer (20 mM NaP, pH 7.5, 20 mM NaCl). After ten column volumes of washing, the nanobody was eluted with 20 mM HEPES, pH 7.5, 2 M $MgCl_2$. The nanobodies were further purified on a HiLoad 16/600 Superdex 75 column in 20 mM HEPES, pH 7.5, 150 mM NaCl, 5% glycerol, concentrated with a 5 kDa cut-off concentrator to 3–10 mg/mL, flash-frozen and stored at −80°C until further use.

### Characterization of nanobody binding to DtpB

The binding of selected nanobodies to DtpB was measured by biolayer interferometry using the Octet RED96 system (Pall ForteBio). The anti-penta-His sensors (Satorius) were incubated in the assay buffer (20 mM HEPES, pH 7.5, 100 mM NaCl, 15 mM imidazole, 5% glycerol, 0.03% DDM) for 10 min. Nanobodies were diluted to 5 μg/ml in the assay buffer, purified DtpB was diluted to 200 nM. All experiments were carried out at 22 °C using the following assay parameters: baseline measurement −180 s, nanobody loading - 300 s, second baseline measurement - 60 s, DtpB association - 300 s, DtpB dissociation - 900 s. Measurements were reference-subtracted and aligned with each other in the Octet Data Analysis HT software v10.0.3.7 (Pall ForteBio), and dissociation constants were determined using a 1:1 binding model.

### Structure determination of DtpB

DtpB and Nb132 were mixed in a 1:1.1 M ratio 1 h prior to crystallization and incubated at 4°C. Crystallization plates were prepared with an automated liquid handler (Mosquito, TTP Labtech) using the sitting drop vapor diffusion technique with a final drop volume of 300 nL (at 1:1, 1:2 and 2:1 (v/v) ratios of protein to precipitant) and as previously described.[43,105,106] A set of crystals were automatically mounted and cryo-cooled through the CrystalDirect Method.[43,44] Peptides were purchased from Sigma-Aldrich, Bachem and GL Biochem (Shanghai) and stocks were prepared by weighing the lyophilized powder using an analytical balance and diluting them in ultrapure water. The initial crystals of the ALA bound structure were obtained from the MemGold2 crystallization screen (Molecular Dimensions) at 19°C. After several rounds of optimization, several DtpB-Nb132 crystals grown in 100 mM HEPES pH 6.7, 30–40% PEG 400 100–300 mM NaCl, 100–300 mM $MgCl_2$, 10 mM ALA, using a protein:precipitant volume ratios of 1:1, 1:2, and 2:1, diffracted X-rays to 2.5 Å. The structures bound to AF, AI, AL, AQ, AV, NV, SL were obtained from crystals grown in the same conditions as for ALA (100 mM HEPES pH 6.7, 30–40% PEG 400 100–300 mM NaCl, 100–300 mM $MgCl_2$, and 10 mM of the respective peptide). Structures bound to KV and MS were obtained from crystals grown in 100 mM MES pH 6.0, 30–40% PEG 400 100–300 mM NaCl, 200 mM $BaCl_2$, and 20 mM of the respective peptides. Structures bound to AFA, AW, AWA were obtained from crystals grown in 100 mM HEPES pH 7.0, 30–40% PEG 400 100–300 mM NaCl, 200 mM $Li_2SO_4$ and 10 mM of the respective peptide. The APF structure was obtained from crystals grown in 100 mM HEPES pH 7.0, 30–40% PEG 400 100–300 mM NaCl, 200 mM $Li_2SO_4$ and 4 mM AFA. Here, once the crystal growth seemed appropriate, the crystals were soaked with 25 mM APF and incubated 1 h at 19°C before harvesting. For all structures, single-crystal monochromatic diffraction datasets of 3600 or 2400 frames were recorded by the rotation method on EIGER 6M detectors with respective oscillation angles of 0.1° and 0.15° at the P13 and P14 beamlines operated by EMBL-Hamburg at the PETRA III storage ring (DESY, Hamburg, Germany). Collected reflections were indexed, integrated and scaled using the program XDS.[95] The unit cell dimensions were roughly: a = 54 Å, b = 124 Å, c = 169 Å, $\alpha = 90^o$, $\beta = 90^o$, $\gamma = 90^o$ and the space group was identified as P 2 21 21. Initial phases were obtained by molecular replacement using the atomic model of DtpA[7] and a nanobody structure as search models in Phaser as part of CCP4i2[96] on the ALA bound dataset. The model was further built manually in Coot[97] and refined in PHENIX[98] and Isolde[99] in iterative cycles. Difference map and OMIT map positive peaks clearly indicated the presence of a ligand within the binding site. Fitting the tripeptide ALA in the density led to better agreement between the experimental data and the atomic model. For the other ligand bound datasets, the latter was repeated with the appropriate peptides and also led to better agreement between the experimental data and the atomic models. Visualisation was performed in ChimeraX.[100]

### Thermal unfolding assay

The NanoDSF method[107] was used to follow the thermal unfolding event of DtpB with the Prometheus NT.48 device (NanoTemper Technologies, Munich, Germany). Here, the fluorescence at 330 (F330) and 350 (F350) nm are recorded over a temperature gradient scan. DtpB was diluted to 0.35 mg/mL with assay buffer (100 mM HEPES, pH 7.5, 150 mM NaCl, 0.03% DDM). All ligands were weighed using a fine scale and dissolved in ultrapure water to a final concentration of 50 mM. 18 μL of DtpB at 0.35 mg/mL was mixed with 2 μL of 50 mM ligand/substrate (5 mM final concentration). Samples were incubated for 10 min at room temperature before loading them with standard capillaries into the Prometheus device. The excitation power was set between 15 and 25%, and the tested temperature range was from 20°C to 85°C. All measurements were done in triplicates and the results were analyzed as described below.

### Determination of $K_D$ from thermal unfolding curves

Thermal unfolding data were exported from the manufacturer's software (PR.ThermControl, v. 2.1.2, NanoTemper GmbH), and the F350/F330 ratio data was analyzed in MoltenProt (v. 0.1.1)[84] using santoro1988 model to determine melting temperature $T_m$. Problematic unfolding curves, e.g., curves with spikes, were omitted from analysis.

Thermal shifts were calculated by subtracting the average $T_m$ of the apo state ($T_m^{apo}$) and normalizing by $T_m^{apo}$ to obtain $\Delta T_m/T_m^{apo}$. The following equation was used to describe the dependence of $\Delta T_m/T_m^{apo}$ on the peptide concentration (titration curves,[55]):

$$\frac{\Delta T_m}{T_m^{apo}}(L) = \frac{-RT_{std}}{E_{a1}} \cdot ln\left(\frac{K_D}{K_D+L}\right) \quad \text{Equation 1}$$

where L is the total ligand concentration (assuming that it is close to the free ligand concentration [L]); R is the universal gas constant; $T_{std}$ is the standard temperature (298.15 K); $E_{a1}$ is activation energy of unfolding for apo state. The equation was derived from Equation 16 in[55] assuming that the affinity of extra-site binding ($K_{DX}$) is much higher than L.

$\Delta T_m$ and $T_m^{apo}$ are determined experimentally, and since R and $T_{std}$ are constants, a non-linear curve fitting procedure can be used to estimate $K_D$ and $E_{a1}$. According to Hall's model, $E_{a1}$ is a property of the apo state of the protein, so it can be a shared parameter between titration curves of several ligands. Once $E_{a1}$ is known for the protein, the $K_D$ of any new ligand can be estimated from $\Delta T_m$ measured at a single ligand concentration:

$$K_D = \frac{L}{exp\left(\frac{\Delta T_m}{T_m^{apo}} \cdot \frac{E_{a1}}{RT}\right) - 1}$$

In this equation, only $\Delta T_m/T_m^{apo}$ and $E_{a1}$ contain experimental errors, so the uncertainty of $K_D$ can be formulated as:

$$\sigma(K_D) = \frac{\frac{\Delta T_m}{T_m^{apo}} \cdot L \cdot E_{a1} \cdot exp\left(\frac{\frac{\Delta T_m}{T_m^{apo}} \cdot E_{a1}}{R \cdot T_{std}}\right) \cdot \sqrt{\left(\frac{\sigma\left(\frac{\Delta T_m}{T_m^{apo}}\right)}{\frac{\Delta T_m}{T_m^{apo}}}\right)^2 + \left(\frac{\sigma(E_{a1})}{E_{a1}}\right)^2}}{R \cdot T_{std} \cdot \left(exp\left(\frac{\frac{\Delta T_m}{T_m^{apo}} \cdot E_{a1}}{R \cdot T_{std}}\right) - 1\right)^2}$$

We first used Equation 1 to determine the $E_{a1}$ and $K_D$ for three peptides with diverse affinities: AA, AF, AQ. If $E_{a1}$ was not a shared parameter, its value fell into a broad range depending on the peptide (348–479 kJ/mol). Furthermore, the standard deviation for $E_{a1}$ obtained from the covariance matrix of the fit parameters was as high as 25–40 kJ/mol. If $E_{a1}$ was a shared parameter between all three titration curves, a consistent fit could be still achieved with $E_{a1}$ 365 ± 8.2 kJ/mol (standard deviation from the fitting procedure). $E_{a1}$ can be also estimated from a thermal unfolding curve of the apo state by applying a two-state irreversible model. This model was previously derived for differential scanning calorimetry (DSC) and NanoDSF data.[84,108] We fit several unfolding curves of DtpB in absence of any ligand using the "irrev" model implemented in MoltenProt (v. 0.1.1), and in this case the $E_{a1}$ was 371 ± 5.7 kJ/mol, which is in good agreement with the value obtained from ligand titration curves.

Fitting of titration curves was performed using symfit (v. 0.5.3),[85] where $E_{a1}$ was shared between all titration curves, while $K_D$ was specific to each curve. The initial value for $E_{a1}$ was set to 326 kJ/mol (corresponds to the consensus value reported by Hall[55]), and initial value for $K_D$ was 1 μM. Depending on the number of curves in the input, either Nelder-Mead minimization algorithm[109] or differential evolution[110] followed by Nelder-Mead minimization was used. To obtain an independent estimate of $E_{a1}$, the unfolding curves of the apo state were fit using irreversible unfolding model "irrev" implemented in MoltenProt.

Raw and processed data as well as the results of intermediate analysis steps were deposited to Zenodo (https://dx.doi.org/10.5281/zenodo.7612000). Python program for shared-parameter fitting of titration (kd.py) is available from Zenodo (https://dx.doi.org/10.5281/zenodo.7611944).

### Determination of $K_D$ using MST

Peptides were diluted in ultrapure water at a highest possible concentration (usually 50–100 mM), and then 2-fold serial dilutions in water were made. DtpB was diluted to 500 nM in 200 mM HEPES, pH 7.5, 300 mM NaCl, 1 mM TCEP, 10% glycerol, 0.06% DDM. Equal volumes of DtpB and peptide stock were mixed and incubated at room temperature 15–60 min. MST was performed using the Monolith NT.LabelFree (NanoTemper GmbH) instrument at room temperature following manufacturer's instructions. Data analysis was performed with MO.Affinity (v. 2.3, NanoTemper GmbH).

### Reconstitution of DtpB into liposomes

DtpB purified in the detergent n-Decyl-β-D-Maltoside (DM; Anatrace) was used for reconstitution into liposomes via the rapid-dilution method. Liposomes were produced by mixing 1- 1-palmitoyl-2-oleoyl-*sn*-glycero-3-phosphoethanolamine (POPE; Avanti polar lipids) and palmitoyl-2-oleoyl-*sn*-glycero-3-phospho-(1′-rac-glycerol) (POPG; Avanti polar lipids) in a 3:1 (w/w) ratio. Chloroform was added to dissolve the lipids and the solvent was removed under vacuum followed by two pentane washes to remove any residual chloroform. The dried lipids were rehydrated in 50 mM potassium phosphate at pH 7.0 (KPi buffer) to reach a final concentration of 20 mg/mL. The lipid mixture was subjected to two freeze-thaw cycles in liquid nitrogen before being stored at −80°C until used. On the day of the reconstitution of DtpB, the appropriate amount of POPE:POPG liposomes were thawed, diluted in 50 mM KPi buffer to a final concentration of 5 mg/mL. Preformed liposomes were obtained by extrusion through a 0.4 μm filter unit (Avanti). DtpB at a

concentration of 0.5 mg/mL was added to the preformed liposomes in a 1:60 (w/w) ratio for pyranine assays or in a 1:10 (w/w) ratio for SURFE$^2$ R assays. For empty liposomes, the same volume of SEC buffer was added instead. The mixture was left to incubate at 4°C for 1h before rapid-dilution into 60 mL of cold 50 mM KPi buffer and subsequent ultracentrifugation at 100,000 g for 1 h at 4°C. The liposomes pellet was resuspended in 50 mM KPi buffer and dialyzed against 50 mM KPi buffer for 72 h at 4°C with intermitted buffer changes. Dialyzed liposomes were harvested for 1h at 60,000 g and 4°C. The liposome pellet was resuspended in 50 mM KPi buffer to reach a final protein concentration of 0.5 mg/mL. Before long-term storage at −80°C, the proteoliposomes were frozen and thawed three-times in liquid nitrogen. Successful reconstitution of DtpB into the liposomes was determined by running the proteoliposomes together with known protein amounts on an SDS-PAGE, followed by densitometry.

### Pyranine-based uptake assays

Proton-coupled uptake of peptides by DtpB, reconstituted 1:60 (w/w) in POPE:POPG liposomes, was indirectly measured using the pH-sensitive dye pyranine (trisodium 8-hydroxypyrene-1,3,6-trisulfonate). On the day of the experiment, proteoliposomes were thawed and the appropriate amount taken for the uptake assay. Here, 5 μg of protein per experiment were used. The proteoliposomes were pelleted by ultracentifugation at 100,000 g for 30 min at 4°C and the pellet was resuspended in internal buffer (5 mM HEPES pH 6.8, 120 mM KCl, 2 mM $MgSO_4$) containing 1 mM pyranine. The mixture was subjected to seven freeze-thaw cycles before being extruded 11-times through at 0.4 μm membrane. After that, the proteoliposomes were centrifuged again at 60,000 g for 25 min at 15°C. The supernatant was discarded and the pellet was resuspended in internal buffer. Residual pyranine was removed by applying the resuspended liposomes onto a pre-equilibrated G-25 spin column (Cytiva). The flow-through was collected and the liposomes pelleted at 60,000 g for 25 min at 15°C. The final proteoliposome pellet was resuspended in internal buffer to reach a final volume of 4 μL per experiment.

All peptide stocks were prepared in distilled water and the pH was adjusted so that the peptide at 2.5 mM final concentration in external buffer (5 mM HEPES pH 6.8, 120 mM NaCl, 2 mM $MgSO_4$) was at 7.0. For the pyranine assay, the prepared proteoliposomes were diluted 1:40 into external buffer in a 96-well black chimney plate (Greiner). Fluorescence of pyranine was measured at excitation wavelengths of 415 nm and 460 nm, and an emission wavelength of 510 nm using a Tecan Spark 20 M Microplate reader. Peptide was added to a final concentration of 2.5 mM and the transport reaction was started via the addition of 1 μM valinomycin. Measurements were performed in a minimum of triplicates with individual measurements performed on separate days.

### Quantification of pyranine transport curves

Data were exported from the plate reader in XLSX format, and the fluorescence signal was calculated by dividing fluorescence at $\lambda_{ex}$ = 460 nm/$\lambda_{em}$ = 510 nm by the fluorescence values at $\lambda_{ex}$ = 415 nm/$\lambda_{em}$ = 510 nm and normalised to 1. Normalized fluorescence curves for empty and DtpB liposomes were first checked to identify the time point when the ligand was added. This was done by finding the largest gaps in the timescale. Empty liposome curve was used to determine the slope of the signal drift by linear fit. The drift was subtracted from the DtpB liposome signal to obtain normalized corrected curves ($RFU_{corr}$), which were then fit either with a straight line or with the exponential equation:

$$RFU_{corr}(t) = RFU_{corr}(t_0) - a \cdot \left(1 - e^{-k_t \cdot (t - t_0)}\right)$$

where $RFU_{corr}(t_0)$ is the signal at the starting time point (fixed value), a is the amplitude of the exponent (parameter bounds from 0 to $RFU_{corr}(t_0)$), τ is the time constant of the observed transport reaction, i.e., time required for the signal to decrease by 1/e of its initial value (parameter bounds from 50 s to infinity), $t_0$ is the starting time of the measurement (parameter bounds from starting time value to ending time value).

Next, the most appropriate model (linear or exponential) for each curve was selected. The residuals of both fits were calculated and assessed for normality using D'Agostino's K-squared test with statistical significance 0.05. Next, the F-statistic was computed for the squared summed residuals (RSS) of both fits:

$$F = \frac{\frac{(RSS_{linear} - RSS_{exp})}{(p_{exp} - p_{linear})}}{\frac{RSS_{exp}}{(n - p_{exp})}}$$

where $RSS_{linear}$, $RSS_{exp}$ – residuals squared and summed for linear or exponential fit, $p_{linear}$, $p_{exp}$ – number of fit parameters (2 parameters in linear fit, 3 parameters in the exponential fit), n – number of data points in a curve. The critical value $F_{crit}$ was computed from F-distribution with 0.01 statistical significance and degrees of freedom ($p_{exp}$ - $p_{linear}$) and (n - $p_{exp}$).

If at least one of the fits had residuals that were not normally distributed, the F-test could not be applied to identify the best model, and the exponential model was selected. If the F-statistic was negative, then the linear model was selected, because the exponent could not fit the curve sufficiently well. If residuals were normally distributed and the F-statistic was above $F_{crit}$, then the exponential model was selected, otherwise the linear model was selected.

For comparison of transport curves from different ligands the instant slope at the starting time point of the measurement slope($t_0$) was computed. In case of exponential fits, this value corresponds to $a/\tau$, while in case of linear fits, slope($t_0$) equals the slope of the line.

All measurements contained a positive transport control using 2.5 mM AA. If the AA curve did not exhibit $\tau$ below 150 s and an amplitude above 0.1, the respective batch of liposomes was considered of poor quality and discarded from subsequent data analysis. Finally, the data were normalized to range 0–1 using slope($t_0$) of buffer-only and AA controls according to the formula:

$$slope(t_0)_{rel} = \frac{slope(t_0)^{peptide} - slope(t_0)^{buffer}}{slope(t_0)^{AA} - slope(t_0)^{buffer}}$$

The slope$(t_0)_{rel}$ from independent experiments (n = 2 or 3) was aggregated using robust equivalents of mean and average (median and median absolute deviation (MAD). This step was necessary to avoid effects of occasional outliers. Curve fitting, data manipulation and visualization were performed in Python 3 with the following modules: numpy,[93] scipy,[94] pandas,[92] matplotlib.[111] Data and code were deposited to Zenodo (https://dx.doi.org/10.5281/zenodo.7612027).

To estimate apparent $K_M$ values, the relative transport rates were loaded into GraphPad Prism (version 9.4.1) and fit to the Michaelis-Menten equation. For $K_i$ measurement for AF, the transport rates at variable AF concentration and fixed AA concentration (2.5 mM) were fit to Hill's equation to determine $IC_{50}$, which was then converted to $K_i$ using the Cheng-Prusoff equation. Curve fitting was performed using scipy function curve_fit.

### Peptide transport assay using SURFE$^2$ R

DtpB reconstituted in 1:10 (w/w) into POPE:POPG liposomes was used for uptake assays using an SSM-based monitoring system. Proteoliposomes were diluted to 1 mg/mL final lipid concentration in 50 mM potassium phosphate buffer pH 7.0 buffer and briefly sonicated. Of this solution, 10 μl were then applied to a prepared SF-N1 sensor (Nanion Technologies) according the manufacturer's instructions. Each prepared sensor was subjected to an initial quality control as recommended by the manufacturer. Sensors that did not meet the criteria of a capacitance of 15–30 nF and conductance of more than 5 nS were not used for subsequent measurements. To monitor proton-influx due to peptide transport, the sensor was washed for 2 s at 200 μL/s with resting buffer (20 mM HEPES pH 6.8, 120 mM KCl, 2 mM $MgSO_4$) containing either Alanine or Glycine (inactivating condition), depending on the peptide used in the measurement. Transport is initiated by a wash with resting buffer containing the substrate at varying concentrations (activating condition). To restore the sensor for the next measurement, it was rinsed again for 2s at 200 μL/s under inactivating conditions. Measurements were performed using DtpB containing liposomes and empty liposomes in triplicates on a single sensor, respectively. The peak currents for the empty liposomes in presence of respective peptides were subtracted from peak currents of DtpB-containing liposomes and loaded into GraphPad Prism (version 9.4.1) to determine apparent $K_M$ and $v_{max}$.

### Flexible docking of di- and tripeptides into DtpB

Flexible peptide docking was accomplished using the Rosetta FlexPepDock protocol[64] (Rosetta version 3.12–2020.08+release. cb1caba). Initial models of all possible di- and tripeptides were generated using the Rosetta BuildPeptide application and then aligned to the N terminus of AF from the experimentally determined DtpB-AF complex (without nanobody) using the Biopython PDB module[86,87] (v. 1.78). These models were minimized (relaxed) using the Rosetta relax application[88] and subjected to the FlexPepDock refinement protocol to obtain 200 docked models. GNU parallel[89] (v. 20180922) in combination with Rosetta's internal parallelization were used to enable efficient processing of the data on a computer cluster.

Peptides with known experimental structures (native models) were used to evaluate the docking performance. First, the residues that build up the native DtpB-peptide interface were identified by performing a contact search within 4 Å distance from all non-hydrogen atoms of the peptide. These residues were used to align the docked models to the native model, and the root-mean-square deviation (RMSD) between the interface residues was recorded (rmsRC). Next, the RMSD between the backbone of the docked and native peptide was calculated (rmsBB_offset). Then the backbone docked peptide was aligned to the backbone of the native peptide and RMSD between the backbone (rmsBB) and side-chains (rmsSC) was calculated. rmsBB and rmsSC characterize the predicted conformation of the peptide; rmsRC characterizes the conformation of DtpB, and rmsBB_offset characterizes the quality of the overall placement of the peptide in the context of DtpB. All alignments and RMSD measurements were done using gemmi[90] (v. 0.4.3). A cutoff for near-native RMSD was set to 1.5 Å.

FlexPepDock protocol uses the "reweighted score" to rank docked models, however, it was observed that in many cases the score does not show a consistent correlation with low RMSD values, in particular, rmsBB_offset. Receiver-operator characteristic (ROC) was used to quantify the performance of the "reweighted score" as a binary classifier between hits (rmsBB_offset below or equal 1.5 Å) and non-hits (rmsBB_offset above 1.5 Å) among all peptides with known native structures. The area under curve (AUC) was 0.68 (AUC for a random classifier is 0.5). Since the FlexPepDock application was designed for peptides that are 5–30 residues long, this result may hint that the default default scoring terms (i.e., weights applied to the components of the Rosetta energy function[65]) are not appropriate for DtpB/peptide interaction. This issue can be tackled by reweighing these terms.[112] To accomplish this in a systematic fashion, Rosetta energy terms for all peptides were extracted using per_residue_energies application and then used to train a regularized logistic regression binary classifier (LR) between hits (rmsBB_offset below or equal 1.5 Å) and non-hits

(rmsBB_offset above 1.5 Å) using scikit-learn[91] (v. 0.24.2). The robustness of the classifier was checked using group k-fold cross-validation (k = 8), corresponding to 2–3 peptides in the test set and 9–10 peptides in the training set. The AUC in this case was 0.79 ± 0.16 (average ±standard deviation). Next, leave-one-out cross-validation (LOOCV) was used to compare the performance LR and "reweighted score". In case of dipeptides LR outperformed "reweighted score" (AUC 0.89 ± 0.08 vs. 0.58 ± 0.14, average ±standard deviation), while for tripeptides "reweighted score" showed better performance (AUC 0.74 ± 0.07 vs. 0.61 ± 0.2, average ±standard deviation). Thus, for dipeptides the final hits were selected by taking the top 10 models ranked by the probability to be a hit reported by LR, and for tripeptides the top 10 models by reweighted_sc were used. Additional code for data manipulation was written in Python (v. 3.7.10) using pandas[92] (v. 1.2.4), numpy[93] (v. 1.20.2) and scipy[94] (v. 1.6.2). Docking results and intermediate steps as well as the code are available from Zenodo (https://dx.doi.org/10.5281/zenodo.7586704). The data analysis workflow is depicted in Figure S12.

### In silico prediction of binding probability

Rosetta energy terms were averaged between the 10 selected hits (see previous section) and used as features for the classifier. The peptides with experimentally determined $K_D$ values were split into two classes: binders ($K_D \leq 1$ mM) and non-binders ($K_D > 1$ mM or no positive $T_m$ shift). The rationale for selecting 1 mM as a cutoff between binders and non-binders stems from our estimate of available peptide concentration in the bacterial growth medium. In case of LB medium, the median concentration of individual amino acids is 2.15 mM,[113] and a significant fraction of amino acids may be in tetrapeptides or larger oligopeptides,[69] which are not bound by DtpB. Thus, in order to be bound and then transported by DtpB, a di- or tripeptide should have a $K_D$ lower than 1–2 mM. From a practical perspective, such binary classification approach (as opposed to regression) is advantageous because peptides that did not produce a significant thermal shift can be still included. This is not possible in regression, because a quantitative estimate of $K_D$ is required. Furthermore, peptides with small $T_m$ shifts have high uncertainty in $K_D$ thus introducing additional noise in the regression. In case of binary classification with logistic regression, such 'uncertain' peptides are being accounted for via yielding a class probability close to 0.5. Finally, the $K_D$ cutoff of 1 mM is close to the median value of the initial peptide $K_D$ dataset (1.38 mM, n = 81), so the experimental binder and non-binder classes are balanced. The features and class assignments were used to train a regularized LR classifier with hyper-parameter tuning using k-fold CV (k = 5) as implemented in scikit-learn[91] (v. 0.24.2). The performance of the classifier was assessed using ROC AUC and k-fold CV (k = 4), which was repeated 20 times using shuffled data. The AUC from this CV was 0.67 ± 0.04 (average ±standard deviation), and AUC using the full dataset was 0.88. $K_D$ cutoffs between 0.5 and 4 mM were also tested, however, ROC AUC characteristics were worse in this case. Model trained on the full dataset was then used to predict binding probability (i.e., probability to have $K_D \leq 1$ mM) for all possible di- and tripeptides. WebLogo[101] (v. 3.7.8) was used to visualize the consensus sequence of di- and tripeptides that have high binding probability ($\geq 0.8$) or low binding probability ($\leq 0.2$). Data manipulation was performed using Python (v. 3.7.10) with pandas[92] (v. 1.2.4), numpy[93] (v. 1.20.2) and scipy[94] (v. 1.6.2). Code and data are available from Zenodo. The data analysis workflow is depicted in Figure S12.